Communication-Aware Adaptive Weights for Consensus-Oriented Distributed KLA-Based LMB Fusion

Author Name[a]

[a]*Affiliation Placeholder, Address Placeholder, City, 000000, State, Country*

ARTICLE INFO

Keywords:
distributed multi-object tracking
labeled multi-Bernoulli
Kullback-Leibler average
adaptive fusion weights
communication-constrained fusion

ABSTRACT

Distributed multi-sensor multi-object tracking over peer-to-peer networks requires fusion rules that remain conservative under unknown cross-correlations while still responding to geometry-induced sensing variation and tiered communication loss. Kullback-Leibler average (KLA), or geometric-average fusion, provides a natural foundation for this setting, but fixed or topology-derived weights cannot express whether a neighbor's posterior is informative, whether its information was actually delivered, or whether its Bernoulli components make decisive existence claims. This paper proposes a communication-aware adaptive weighting family for consensus-oriented distributed KLA-based LMB fusion. The method uses a covariance, realized-link, and existence-confidence backbone, then separates spatial and existence weighting so that a weak structure-aware correction regularizes the spatial branch and an FID-FIA information-geometric cue can be reserved for existence fusion. The evaluation uses an eight-sensor dual-formation scenario with finite fields of view and tiered heterogeneous packet loss, and treats network disagreement, local tracking accuracy, and filter/fusion runtime as complementary evidence. The results show that the proposed operating modes improve consensus over fixed and dynamic-weight baselines without collapsing conventional local metrics: the Balanced mode is the lower-overhead position-oriented setting, whereas the Cardinality-critical mode uses additional information-geometric computation to obtain the strongest target-number agreement. The study therefore frames adaptive distributed GA-LMB fusion as an explicit spatial-versus-cardinality-versus-cost tradeoff rather than as a single fixed weighting recipe.

1. Introduction

Distributed multi-sensor multi-object tracking is a core capability in surveillance, autonomous systems, and cooperative perception. The difficult case considered in this paper is not centralized multi-view fusion with a reliable communication backbone, but peer-to-peer tracking in which each node has partial spatial coverage, local posterior quality varies over time, and communication is affected by bandwidth limits, packet drops, or temporary outages (Zhang, Chen, Song and Yu, 2016). Random finite set (RFS), finite-set statistics (FISST), and labeled-RFS filters provide the Bayesian state representation needed for this setting because they model target number, kinematic state, and identity in a single multi-object posterior (Mahler, 2003; Vo and Ma, 2006; Mahler, 2007; Vo, Vo and Phung, 2014; Reuter, Vo, Vo and Dietmayer, 2014). The fusion problem then becomes a second-stage question: after each node has formed a local labeled posterior, how should neighboring posteriors be combined when field-of-view geometry, posterior uncertainty, and packet delivery all vary across the peer-to-peer network?

Within the labeled-RFS family, GLMB and LMB filters offer a practical balance between multi-object modeling fidelity and computational tractability (Vo et al., 2014; Reuter et al., 2014; Vo, Vo and Beard, 2019; Mahler, 2019; Vo, Vo, Nguyen and Shim, 2024). For distributed fusion, Kullback-Leibler average (KLA), equivalently geometric-average or generalized covariance-intersection fusion, is attractive because it gives a conservative logarithmic-opinion-pool rule when unknown cross-correlations make exact Bayesian fusion unsafe (Julier and Uhlmann, 1997; Mahler, 2000; Battistelli and Chisci, 2014; Hlinka, Sluciak, Hlawatsch and Rupp, 2014; Üney, Clark and Julier, 2013; Bandyopadhyay and Chung, 2018). This makes KLA-based LMB fusion a natural backbone for peer-to-peer multi-object tracking. The weakness is not the conservative fusion principle itself, but the usual assumption that the fusion weights can be fixed, uniform, or derived primarily from network topology.

Choosing these weights is therefore the central practical bottleneck. Fixed Metropolis, uniform, or degree-based consensus weights (Xiao, Boyd and Lall, 2005) are easy to deploy, but they cannot distinguish a precise local posterior

author@example.com (A. Name)
ORCID(s):

from an ambiguous one, and they cannot tell whether a nominally useful neighbor has actually delivered useful information at the current time step. Existing adaptive-weight ideas address parts of this issue. Centralized and complementary LMB fusion studies adapt sensor contributions under heterogeneous fields of view (Wang, Gostar, Rathnayake, Xu, Bab-Hadiashar and Hoseinnezhad, 2018; Gostar, Rathnayake, Tennakoon, Bab-Hadiashar, Battistelli, Chisci and Hoseinnezhad, 2021; Klupacs, Gostar, Bab-Hadiashar and Hoseinnezhad, 2023); distributed RFS fusion has also begun to assign weights to local subdensity components in peer-to-peer networks (Zheng, Shi, Yang and Cai, 2026). Information-geometric and Fisher-information weighting further show that local information content can be a principled weighting signal (Cao and Zhao, 2026; Li, Li, Li and Miao, 2026). Existing adaptive weighting rules are often organized around a dominant cue, such as graph topology, local information content, detectability, or Fisher-information-based target separability. The missing piece is not another isolated cue, but a factorized and branch-aware weighting framework for communication-constrained peer-to-peer GA-LMB fusion that can systematically evaluate and combine complementary reliability cues rather than forcing a single scalar signal to govern the whole posterior.

This general gap becomes especially visible in the two-part structure of the LMB posterior: a single scalar reliability score is unlikely to characterize both the spatial accuracy of a Bernoulli component and the decisiveness of its existence probability. In GA-LMB fusion, a node may provide a well-localized state estimate while remaining uncertain about whether the corresponding object exists, or it may provide decisive existence evidence whose spatial density is still comparatively diffuse.

Motivated by this gap, this paper proposes a communication-aware adaptive fusion-weight allocation family for distributed GA-LMB fusion. The design first builds a shared quality backbone from covariance quality, realized link quality, and existence confidence. It then separates the spatial and existence branches of the LMB fusion rule: a weak structure-aware correction regularizes spatial fusion without letting topology dominate the posterior-quality terms, while an FID-FIA-informed refinement imports the Fisher-information-accumulation idea of Cao and Zhao (Cao and Zhao, 2026) only into the Bernoulli existence branch. Figure 1 summarizes this peer-to-peer update, communication, and branch-decoupled fusion workflow. This leads to two operating modes rather than a single claimed optimum: a Balanced mode for spatial/position-sensitive deployment and a Cardinality-critical mode for target-number agreement.

The evaluation follows the same logic. The main experiment is an eight-sensor dual-formation tracking scenario under tiered heterogeneous packet loss, chosen to stress geometry/FOV-induced posterior-quality variation and realized communication heterogeneity. Network-level disagreement metrics are the primary outcome because the method is designed to reduce disagreement in the fused multi-sensor picture; truth-referenced local metrics and filter/fusion runtime are reported as safeguards against hidden accuracy or deployment-cost regressions. The results support the intended tradeoff: communication-aware adaptive weighting improves consensus over fixed and dynamic-weight baselines, the covariance/link/existence backbone explains most of the low-overhead gain, and the existence-only FID-FIA refinement is most useful when cardinality agreement is the dominant operational risk.

The main contributions of this paper are as follows.

1. We propose a communication-aware adaptive fusion-weight allocation family for distributed KLA-based GA-LMB fusion under communication constraints.
2. We introduce a factorized quality model that combines posterior covariance, realized link quality, and existence confidence, together with branch-specific refinements: weak structure-aware modulation for spatial fusion and FID-FIA-informed information geometry for existence fusion.
3. We show in a tiered-loss dual-formation eight-sensor scenario, with ideal-communication, communication-sensitivity, and runtime support, that the method improves network-level consensus over fixed-weight and partial-factor baselines while exposing the low-overhead balanced and higher-cost cardinality-critical operating choices.

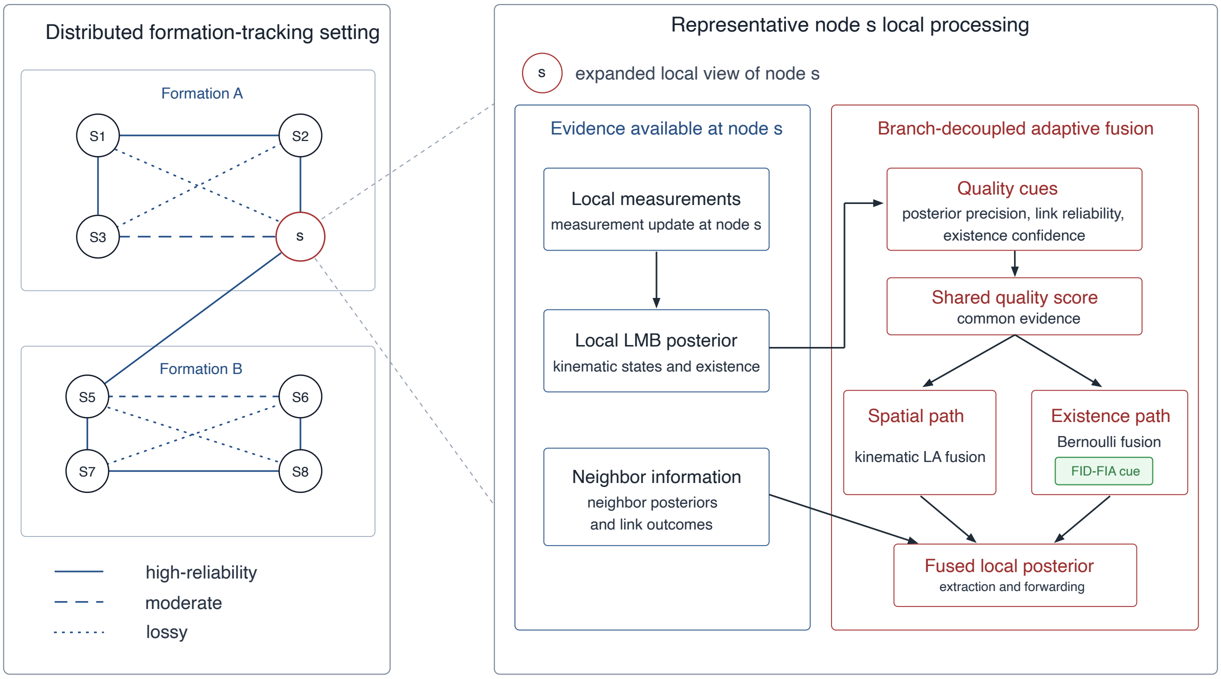

Figure 1: System overview of the distributed eight-sensor tracking setting, including local measurement update, neighborhood communication, branch-decoupled adaptive GA-LMB fusion, and an FID-FIA cue applied only on the existence-weight path.

2. Related Work

The most relevant prior work falls into three groups: labeled-RFS filtering, conservative distributed density fusion, and adaptive or weighted RFS/LMB fusion. The first group provides the statistical state representation. Mahler's RFS/FISST formulation and PHD recursion establish the set-valued Bayesian foundation (Mahler, 2003, 2007; Vo and Ma, 2006), while GLMB and LMB filters add labeled multi-object posteriors suitable for track identity and tractable implementation (Vo et al., 2014; Reuter et al., 2014; Vo et al., 2019; Mahler, 2019; Vo et al., 2024). The present paper does not modify this filtering family; it uses LMB posteriors as the objects to be fused.

The second group concerns conservative fusion when cross-correlations are unknown. Covariance intersection, Kullback-Leibler averaging, geometric-average fusion, and inverse-covariance-intersection variants provide density-combination rules that avoid overconfident fusion under unknown dependence (Julier and Uhlmann, 1997; Mahler, 2000; Battistelli and Chisci, 2014; Hlinka et al., 2014; Uhlmann, 2003; Noack, Sijs, Reinhardt and Hanebeck, 2017; Zhang, 2020). Related logarithmic-opinion-pool and exponential-mixture formulations connect the same idea to distributed Bayesian fusion and consensus (Üney et al., 2013; Bandyopadhyay and Chung, 2018). In the labeled-RFS setting, GCI/KLA-style fusion has been extended to multi-Bernoulli and labeled-RFS densities, including work on label mismatch and peer-to-peer resilience (Wang, Yi, Hoseinnezhad, Li, Kong and Yang, 2017; Li, Yi, Hoseinnezhad, Battistelli, Wang and Kong, 2018; Gao, Battistelli and Chisci, 2023). These studies justify the GA/KLA backbone used here, but they leave open how the weights should respond to time-varying posterior quality and realized communication quality.

The third group is most directly related to adaptive weighting. Centralized multiple-view LMB fusion shows that information-aware weights can help when sensors have limited or different fields of view (Wang et al., 2018; Gostar et al., 2021). Klupacs et al. take a complementary-fusion view, fusing label-associated Bernoulli densities and tuning sensor contribution by object detectability (Klupacs et al., 2023). Consensus-based distributed LMB tracking has also explored automatic track-level weight evolution for different fields of view (Shen, Dong, Jing and Leung, 2022). Recent peer-to-peer RFS work moves still closer to the present setting by factorizing local RFS densities into subdensities and assigning local weighting coefficients before GA or AA fusion under consensus or flooding protocols (Zheng et al., 2026). The closest information-geometric GA/KLA precedent is Cao and Zhao's distributed heterogeneous labeled-RFS method, which uses Fisher information distance and Fisher information accumulation (FID-FIA) to assign adaptive scalar fusion weights under heterogeneous sensing architectures (Cao and Zhao, 2026). In parallel, AA density fusion and related minimum-information-loss or cardinality-consistent formulations clarify the theoretical contrast between AA and GA fusion (Li, Yan, Da and Fan, 2024; Li, Corchado and Sun, 2017; Gao, Battistelli and Chisci, 2020; Üney, Houssineau, Delande, Julier and Clark, 2019). A recent TechRxiv preprint in the AA-density-fusion series further studies Fisher-information-proportional weights for time-aligned asynchronous multi-rate filters and reports their use in both AA and GA fusion (Li et al., 2026). These works collectively support adaptive information-aware weighting, but they do not combine realized packet delivery, LMB branch decoupling, and existence-only information-geometric refinement in the synchronous peer-to-peer GA-LMB packet-loss setting studied here.

The present method builds on these precedents but narrows the design question. Rather than switching to complementary union, replacing the whole GA-LMB posterior by one FID-FIA scalar weight, or solving asynchronous time alignment, it retains the distributed KLA-based GA-LMB operator and changes the weight-allocation mechanism around it. The backbone uses posterior concentration and realized packet delivery, the Bernoulli side adds existence decisiveness, and the optional information-geometric cue is confined to existence fusion. This distinction is central to the paper's operating-mode claim: the Balanced mode is intended for low-overhead spatial consensus, whereas the Cardinality-critical mode spends additional FID-FIA computation when target-number agreement is more important than the local-RMSE tradeoff.

Communication constraints provide the deployment motivation for the weighting problem. Bandwidth limits, delays, asynchronous updates, and packet losses are central design factors in distributed fusion over sensor networks (Zhang et al., 2016). The present work focuses on the packet-delivery side of this problem: realized communication quality enters directly into the fusion weights, rather than appearing only as a nominal graph topology or an external simulation condition.

Against this background, the contribution of this paper is deliberately narrower than a broad new fusion framework. We stay within distributed KLA-based GA-LMB fusion, replace fixed or topology-only weighting with a covariance, realized-link, and existence-confidence backbone, and add branch-specific refinements for spatial and existence fusion. This positioning differentiates the method from both fixed-weight KLA fusion and adaptive scalar-weight formulations that do not explicitly decouple the LMB branches under heterogeneous packet loss.

3. Problem Formulation

Consider a distributed sensor network with sensor index set $\mathcal{V} = \{1, \dots, S\}$ operating over discrete time $k \in \{1, \dots, T\}$. The multi-object state at time k is modeled as a labeled random finite set X_k, and each sensor $s \in \mathcal{V}$ acquires a local measurement set $Z_k^{(s)}$. Following the standard RFS/FISST and labeled-RFS formulation, each node maintains an LMB approximation to the local multi-object posterior, whose Bernoulli components jointly encode target existence, kinematic state, and identity (Mahler, 2003, 2007; Vo et al., 2014; Reuter et al., 2014; Vo et al., 2019). The objective of the distributed fusion layer is to combine neighboring local posteriors into a consensus-quality posterior without assuming that cross-correlations among nodes are known.

3.1. Local LMB posterior model

Let $\mathcal{N}_s \subseteq \mathcal{V}$ denote the communication neighborhood of sensor s, including the node itself. In the setting considered here, $\mathcal{N}_s$ is induced by a communication graph derived from sensor geometry and communication range, and each node performs local fusion only over the posteriors available inside this neighborhood. After local prediction, measurement update, and data association, node $j \in \mathcal{N}_s$ produces a measurement-updated LMB posterior

$$\pi_k^{(j)} = \left\{ \left(r_{k,i}^{(j)}, p_{k,i}^{(j)}(\cdot, \ell_i) \right) \right\}_{i=1}^{M_k}, \tag{1}$$

where $r_{k,i}^{(j)} \in [0, 1]$ is the Bernoulli existence probability and $p_{k,i}^{(j)}$ is the associated spatial density for label ℓ_i.

3.2. Communication-constrained delivery

Before local measurements are used by a neighboring fusion node, they pass through a communication layer with three possible effects:

1. a global or per-step bandwidth budget that limits how many measurements can be delivered,
2. a stochastic packet-drop model on each sensor link,
3. optional node-level outages.

Accordingly, the posterior that reaches a neighbor is influenced not only by the underlying sensing quality, but also by whether measurements were actually delivered. Let $m_k^{(j)} \in \{0, 1\}$ denote the availability indicator of node j at time k, and let $\rho_k^{(j)} \in [0, 1]$ denote the realized link-quality statistic derived from delivered-versus-dropped packets.

3.3. Distributed KLA fusion objective

Because the cross-correlation structure among neighboring LMB posteriors is unknown, exact Bayesian fusion is generally unavailable. We therefore adopt a conservative Kullback-Leibler-average or geometric-average fusion rule. At node s, the target fused density over the neighborhood $\mathcal{N}_s$ is

$$\bar{\pi}_k^{(s)}(X) \propto \prod_{j \in \mathcal{N}_s} \left(\pi_k^{(j)}(X)\right)^{\omega_{k,s}^{(j)}}, \qquad \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{(j)} = 1, \qquad \omega_{k,s}^{(j)} \geq 0, \tag{2}$$

where $\omega_{k,s}^{(j)}$ is the fusion weight assigned by node s to neighbor j at time k (Battistelli and Chisci, 2014; Hlinka et al., 2014). With fixed-weight fusion, $\omega_{k,s}^{(j)}$ is often chosen as a uniform or Metropolis-type consensus weight (Xiao et al., 2005). The central problem studied here is how to choose these weights adaptively so that they reflect both posterior quality and communication quality.

In the GA-LMB fusion rule considered here, the spatial and existence parts of each Bernoulli component are fused separately after moment matching. This separation makes it natural to ask whether the same scalar weight should govern both spatial fusion and existence fusion.

3.4. Adaptive weight-allocation objective

The adaptive weighting design starts from a shared quality backbone. For node s and neighbor $j \in \mathcal{N}_s$, define the unnormalized score

$$\tilde{\omega}_{k,s}^{(j)} = m_k^{(j)} \cdot q_{\mathrm{cov},k}^{(j)} \cdot q_{\mathrm{link},k}^{(j)} \cdot q_{\mathrm{exist},k}^{(j)}, \tag{3}$$

where $m_k^{(j)}$ is the availability mask, $q_{\mathrm{cov},k}^{(j)}$ is a covariance-quality score, $q_{\mathrm{link},k}^{(j)}$ is a realized link-quality score, and $q_{\mathrm{exist},k}^{(j)}$ is an existence-confidence score. The normalized fusion weight is then

$$\omega_{k,s}^{(j)} = \frac{\tilde{\omega}_{k,s}^{(j)}}{\sum_{u \in \mathcal{N}_s} \tilde{\omega}_{k,s}^{(u)}}. \tag{4}$$

3.5. Decoupled and weakly structure-aware weighting

The branch-decoupled fusion rule does not force the spatial and existence branches to share exactly the same dynamics. Instead, it forms branch-specific scores,

$$\tilde{\omega}_{k,s}^{x,(j)} = \mathrm{blend}\left(\tilde{\omega}_{k,s}^{(j)}, \tilde{\omega}_{k,s,\mathrm{sp}}^{(j)}, \eta_x\right), \qquad \tilde{\omega}_{k,s}^{r,(j)} = \mathrm{blend}\left(\tilde{\omega}_{k,s}^{(j)}, \tilde{\omega}_{k,s,\mathrm{ex}}^{(j)}, \eta_r\right), \tag{5}$$

and then applies only a weak graph-aware correction,

$$\tilde{\omega}_{k,s}^{x,(j)} \leftarrow \tilde{\omega}_{k,s}^{x,(j)} \cdot \left(\xi_{k,s}^{x,(j)}\right)^{\gamma_x}, \qquad \tilde{\omega}_{k,s}^{r,(j)} \leftarrow \tilde{\omega}_{k,s}^{r,(j)} \cdot \left(\xi_{k,s}^{r,(j)}\right)^{\gamma_r}. \tag{6}$$

Here $\xi_{k,s}^{x,(j)}$ and $\xi_{k,s}^{r,(j)}$ are local structure priors derived from neighborhood overlap and communication reliability. The existence-side exponent is chosen much smaller than the spatial-side exponent, $\gamma_r \ll \gamma_x$, so topology acts as a mild prior layered on top of posterior-quality and communication-quality signals rather than as the primary source of fusion weights.

3.6. Evaluation target

The aim of the above formulation is not only to improve each node's local tracking output, but also to improve agreement among nodes. This distinction is central in the present problem. Under unknown cross-correlations and communication losses, distributed fusion is useful only if neighboring nodes move toward a shared multi-sensor picture rather than drifting toward mutually inconsistent local beliefs.

Accordingly, the evaluation uses two complementary metric layers:

- primary network-level disagreement metrics, which quantify consensus error among the fused outputs of different sensors,
- secondary truth-referenced local tracking metrics, which verify that consensus-error reductions are not obtained by materially degrading conventional tracking accuracy.

Let $\hat{X}_k^{(s)}$ denote the extracted multi-object estimate at sensor s and time k, and let $\hat{n}_k^{(s)} = |\hat{X}_k^{(s)}|$ denote its estimated cardinality. The paper defines the following network-level disagreement metrics:

$$D_{\mathrm{OSPA}}(k) = \frac{2}{S(S-1)} \sum_{1 \le a < b \le S} d_{\mathrm{OSPA}}\left(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}\right), \tag{7}$$

$$D_{\mathrm{loc}}(k) = \frac{2}{S(S-1)} \sum_{1 \le a < b \le S} d_{\mathrm{loc}}\left(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}\right), \tag{8}$$

$$D_{\mathrm{card}}(k) = \frac{1}{S} \sum_{s=1}^{S} \left| \hat{n}_k^{(s)} - \mathrm{median}\left(\hat{n}_k^{(1)}, \dots, \hat{n}_k^{(S)}\right) \right|, \tag{9}$$

where d_{OSPA} is the pairwise OSPA distance between two estimated finite sets and d_{loc} is the matched localization disagreement obtained after Hungarian matching on the common extracted objects (Schuhmacher, Vo and Vo, 2008; Ristic, Vo, Clark and Vo, 2011; Kuhn, 1955). In the paper these are reported as OSPA consensus error, matched localization disagreement, and cardinality dispersion, respectively.

These three quantities are task-driven network-level disagreement measures rather than standard field-wide tracking benchmarks. Their building blocks are standard or conventional: OSPA and related GOSPA-style distances are mainstream finite-set tracking metrics, Hungarian matching is the standard assignment primitive behind many tracking scores, and pairwise dispersion is the usual way to quantify disagreement in consensus problems (Schuhmacher et al., 2008; Rahmathullah, Garcia-Fernandez and Svensson, 2017; Ristic et al., 2011; Xiao et al., 2005). What is specific to this paper is the aggregation target: instead of comparing each sensor only with ground truth, the metrics compare the post-fusion outputs of different nodes. They are introduced because the present problem is distributed fusion under heterogeneous communication, where the main question is whether neighboring nodes move toward a common fused picture. For that reason, the paper does not use them as replacements for conventional truth-referenced tracking metrics, and it pairs the network-level disagreement metrics with the secondary local metrics listed below.

The secondary local metrics remain the standard truth-referenced tracking quantities:

- local E-OSPA,
- local RMSE,
- local cardinality error.

The evaluation logic is therefore asymmetric by design: the method is primarily successful if it reduces network-level disagreement, while the local metrics are used as a safeguard to show that this gain is not purchased by a clear loss in conventional tracking quality.

4. Communication-Aware Adaptive KLA Fusion

This section describes the adaptive fusion rule used in the distributed GA-LMB filter. The starting point is a conservative Kullback-Leibler-average or geometric-average fusion strategy for neighboring LMB posteriors under unknown cross-correlations (Battistelli and Chisci, 2014; Hlinka et al., 2014). The main methodological question is not how to redesign the underlying labeled-RFS filter family, but how to allocate fusion weights so that they reflect posterior quality and realized communication quality in a communication-constrained sensor network. A variational and information-geometric interpretation of the branch-decoupled update is given in Appendix C.

4.1. Baseline GA-LMB fusion

At node s, let $\mathcal{N}_s$ denote the local communication neighborhood and let $\pi_k^{(j)}$ be the measurement-updated LMB posterior provided by node $j \in \mathcal{N}_s$. The baseline distributed fusion target is the weighted geometric average

$$\bar{\pi}_k^{(s)}(X) \propto \prod_{j\in\mathcal{N}_s} \left(\pi_k^{(j)}(X)\right)^{\omega_{k,s}^{(j)}}, \qquad \sum_{j\in\mathcal{N}_s} \omega_{k,s}^{(j)} = 1, \qquad \omega_{k,s}^{(j)} \geq 0. \tag{10}$$

After moment projection, the Gaussian approximation of the ith Bernoulli spatial density at node j is

$$p_{k,i}^{(j)}(x) \approx \mathcal{N}\left(x; v_{k,i}^{(j)}, T_{k,i}^{(j)}\right). \tag{11}$$

Using spatial fusion weights $\omega_{k,s}^{x,(j)}$, the fused spatial density is obtained in canonical form:

$$K_{k,i}^{(s)} = \sum_{j\in\mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1}, \qquad h_{k,i}^{(s)} = \sum_{j\in\mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1} v_{k,i}^{(j)}, \tag{12}$$

$$\Sigma_{k,i}^{(s)} = \left(K_{k,i}^{(s)}\right)^{-1}, \qquad \mu_{k,i}^{(s)} = \Sigma_{k,i}^{(s)} h_{k,i}^{(s)}. \tag{13}$$

Using existence fusion weights $\omega_{k,s}^{r,(j)}$, the Bernoulli existence probability is fused by the corresponding weighted geometric rule:

$$r_{k,i}^{(s)} = \frac{\eta_{k,i}^{(s)} \prod_{j\in\mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}{\eta_{k,i}^{(s)} \prod_{j\in\mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}} + \prod_{j\in\mathcal{N}_s} \left(1 - r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}, \tag{14}$$

where $\eta_{k,i}^{(s)}$ is the Gaussian normalization term induced by the fused spatial density.

4.2. Factorized adaptive weight backbone

The adaptive design begins with a shared quality backbone. For node s and neighbor $j \in \mathcal{N}_s$, define the raw score

$$\tilde{\omega}_{k,s}^{(j)} = m_k^{(j)} \cdot q_{\mathrm{cov},k}^{(j)} \cdot q_{\mathrm{link},k}^{(j)} \cdot q_{\mathrm{exist},k}^{(j)}, \tag{15}$$

where $m_k^{(j)} \in \{0, 1\}$ is the availability mask, $q_{\mathrm{cov},k}^{(j)}$ is a covariance-quality score, $q_{\mathrm{link},k}^{(j)}$ is a realized link-quality score, and $q_{\mathrm{exist},k}^{(j)}$ is an existence-confidence score. The normalized weight is

$$\bar{\omega}_{k,s}^{(j)} = \frac{\tilde{\omega}_{k,s}^{(j)}}{\sum_{u\in\mathcal{N}_s} \tilde{\omega}_{k,s}^{(u)}}. \tag{16}$$

This factorization focuses on the three factors that are most directly aligned with the design objective: posterior concentration through covariance, communication reliability through delivered-versus-dropped packets, and existence decisiveness through Bernoulli existence probabilities. Figure 2 shows how this shared backbone feeds the decoupled spatial and existence-weight paths used by the two operating modes.

4.3. Covariance and link-quality terms

For node j, let $T_{k,i}^{(j)}$ be the moment-matched covariance of the ith Bernoulli component. The covariance-quality score is defined as

$$q_{\mathrm{cov},k}^{(j)} = \frac{1}{\epsilon + \frac{1}{M_k} \sum_{i=1}^{M_k} \mathrm{tr}\left(T_{k,i}^{(j)}\right)}, \tag{17}$$

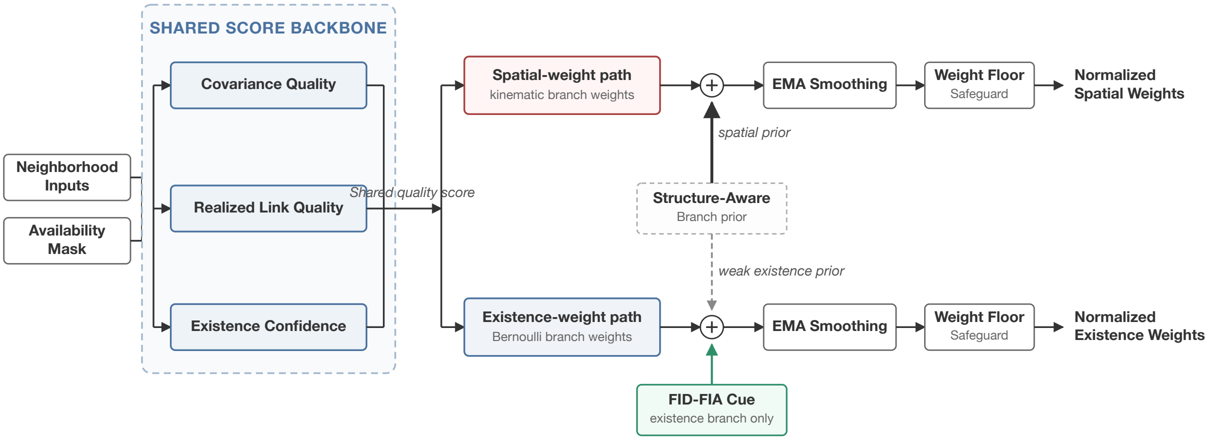

Figure 2: Adaptive fusion-weight factorization, showing the shared quality backbone, decoupled spatial and existence-weight paths, weak structure-aware branch prior, and an FID-FIA cue applied only to the existence branch.

so that a node with smaller posterior spread receives a larger score.

The link-quality term is derived from realized communication outcomes rather than nominal topology. Let $d_k^{(j)}$ and $\ell_k^{(j)}$ denote the counts of delivered and dropped measurements, respectively. The realized link-quality score is defined as

$$q_{\mathrm{link},k}^{(j)} = \frac{d_k^{(j)}}{d_k^{(j)} + \ell_k^{(j)}}, \tag{18}$$

which directly reflects how reliably that node's information is reaching its neighbors.

4.4. Existence-confidence weighting

To capture the missing existence-related dimension, the method computes a certainty score from Bernoulli existence probabilities. For the ith Bernoulli component at node j,

$$c_{k,i}^{(j)} = \left|2r_{k,i}^{(j)} - 1\right|. \tag{19}$$

The per-node weighted existence confidence is then

$$\bar{c}_k^{(j)} = \frac{\sum_{i=1}^{M_k} r_{k,i}^{(j)} c_{k,i}^{(j)}}{\epsilon + \sum_{i=1}^{M_k} r_{k,i}^{(j)}}. \tag{20}$$

The final existence-confidence score is mapped into a bounded positive factor:

$$q_{\mathrm{exist},k}^{(j)} = \lambda_{\min} + (1 - \lambda_{\min}) \left(\bar{c}_k^{(j)}\right)^{p_e}, \tag{21}$$

where $\lambda_{\min} \in (0, 1]$ and $p_e > 0$ are tunable parameters. Figure 3 visualizes this bounded Bernoulli-level mapping and representative node-level aggregation effects.

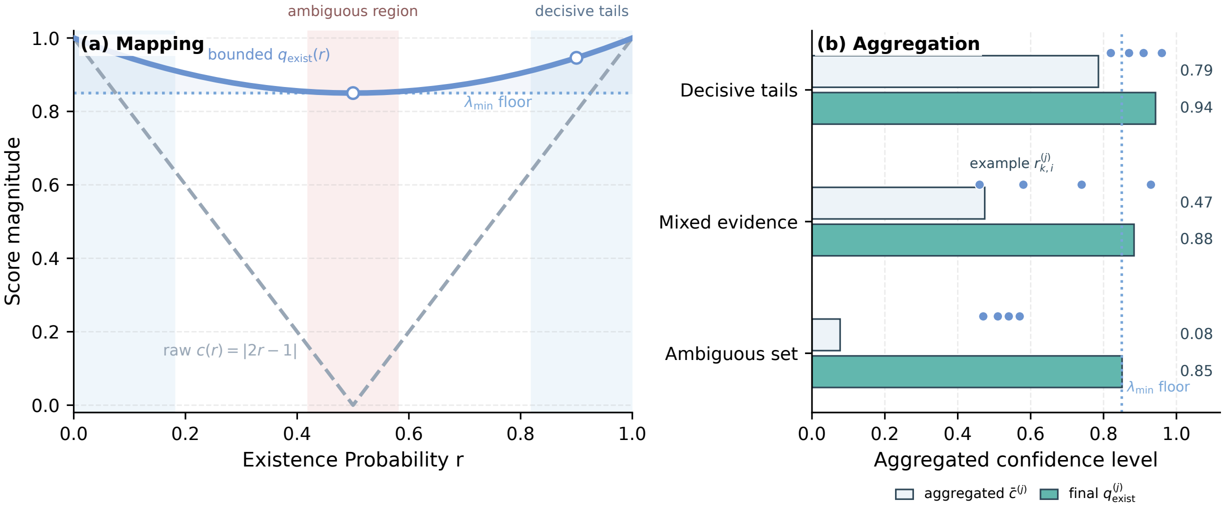

Figure 3: Existence-confidence weighting used in the adaptive rule: (a) the Bernoulli-level decisiveness mapping and bounded score curve, and (b) representative node-level aggregation examples showing how decisive posterior sets produce larger $\bar{c}^{(j)}$ and $q_{\text{exist}}^{(j)}$.

4.5. Decoupled spatial and existence scores

The proposed method does not force the same adaptive score to govern spatial fusion and existence fusion. This separation is structurally motivated by the Bernoulli/LMB form itself: existence and spatial terms enter the fused posterior through different update blocks, even though the formal variational interpretation is deferred to Appendix C. The method therefore constructs branch-specific dedicated scores:

$$\tilde{\omega}_{k,s,\text{sp}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{cov},k}^{(j)}\right)^{\alpha_x} \cdot \left(q_{\text{link},k}^{(j)}\right)^{\beta_x}, \tag{22}$$

$$\tilde{\omega}_{k,s,\text{ex}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{link},k}^{(j)}\right)^{\beta_r} \cdot \left(q_{\text{exist},k}^{(j)}\right)^{\alpha_r}. \tag{23}$$

The final branch scores are obtained through geometric interpolation:

$$\tilde{\omega}_{k,s}^{x,(j)} = \left(\tilde{\omega}_{k,s}^{(j)}\right)^{1-\eta_x} \left(\tilde{\omega}_{k,s,\text{sp}}^{(j)}\right)^{\eta_x}, \qquad \tilde{\omega}_{k,s}^{r,(j)} = \left(\tilde{\omega}_{k,s}^{(j)}\right)^{1-\eta_r} \left(\tilde{\omega}_{k,s,\text{ex}}^{(j)}\right)^{\eta_r}. \tag{24}$$

The point of this decoupling is not to create two unrelated methods, but to acknowledge that spatial consensus and existence consensus respond differently to weight perturbations. It also determines how Fisher-information-based cues are used in this paper. The information-geometric construction of Cao and Zhao is a natural scalar indicator of local target-separability and posterior informativeness, and in our experiments it is especially effective for cardinality and existence decisions (Cao and Zhao, 2026). Related multi-rate average-fusion work also supports Fisher information as a principled local information measure after time alignment (Li et al., 2026). However, applying the same Fisher-type scalar weight to the whole fused posterior can trade away spatial RMSE in the present GA-LMB setting. In the log-utility view developed in Appendix C, a single FID-FIA scalar-weight baseline is a constrained case that forces the same Fisher-separability utility into both the Gaussian spatial barycenter and the Bernoulli existence pool. The branch-decoupled formulation relaxes this constraint: the communication-aware covariance/link-quality branch continues to govern spatial fusion, while the FID-FIA signal is reserved for the existence branch where its empirical strength is most aligned with the fused Bernoulli variable.

4.6. FID-FIA existence refinement

The Cardinality-critical mode therefore adds an information-geometric refinement only to the existence branch. This refinement follows the FID-FIA intuition of accumulating pairwise Fisher-information distance between locally represented targets, but it is not used as a replacement scalar fusion rule. For each node j, the method first computes a target-pair accumulation score $a_k^{(j)}$ from the local measurement-updated Bernoulli components. Each Bernoulli component is moment-projected, target pairs are weighted by their existence probabilities, and the pairwise distance is accumulated along the line segment between the two projected states using the local Fisher metric induced by the sensor's detection probability and measurement covariance. The score is then normalized over the available local neighborhood,

$$\hat{a}_{k,s}^{(j)} = \frac{a_k^{(j)}}{\epsilon + \max_{u \in \mathcal{N}_s} a_k^{(u)}}, \qquad 0 \le \hat{a}_{k,s}^{(j)} \le 1. \tag{25}$$

This normalized accumulation is mapped into a bounded existence-only modulation factor,

$$q_{\mathrm{fid},k,s}^{(j)} = \rho_{\min} + (1 - \rho_{\min}) \left(\hat{a}_{k,s}^{(j)} \right)^{p_f}. \tag{26}$$

Only the existence score is modified:

$$\tilde{\omega}_{k,s}^{r,(j)} \leftarrow \tilde{\omega}_{k,s}^{r,(j)} \left(q_{\mathrm{fid},k,s}^{(j)} \right)^{\gamma_f}. \tag{27}$$

The spatial score $\tilde{\omega}_{k,s}^{x,(j)}$ is left unchanged. In the main experiment, this branch uses a strong existence-only modulation and disables temporal smoothing and minimum-weight flooring for the existence weights, while leaving the spatial branch's structure-aware stabilized weights unchanged. Equivalently, for positive $q_{\mathrm{fid},k,s}^{(j)}$, the update adds $\gamma_f \log q_{\mathrm{fid},k,s}^{(j)}$ to the existence-branch utility and adds nothing to the spatial utility; the zero-floor setting used in the main experiment is the boundary case that permits a zero instantaneous existence weight.

4.7. Recommended operating modes

The resulting method family supports two practical operating modes rather than a single mandatory setting. The *Balanced mode* uses the three-factor adaptive backbone, decoupled spatial and existence scores, weak structure-aware refinement, and the standard temporal stabilization, but does not apply the final FID-FIA existence modulation. This mode is recommended when spatial accuracy, RMSE stability, runtime, or energy budget is the primary operational concern.

The *Cardinality-critical mode* adds the FID-FIA cue only to the existence branch. This mode is recommended when the cost of target-number disagreement, missed tracks, or false support is higher than a modest local-RMSE and runtime tradeoff. Both modes share the same communication-aware backbone and the same branch-decoupled fusion algebra; they differ only in whether the final information-geometric cue is activated on the Bernoulli existence path. This framing is important because the experiments show a real operating tradeoff: the backbone is often the stronger low-overhead position-oriented choice, whereas the Cardinality-critical mode is the stronger cardinality-consensus choice with substantially higher information-geometric computation.

4.8. Weak structure-aware refinement and stabilization

After branch decoupling and before the final FID-FIA existence modulation, the method applies a weak graph-aware correction through local structure priors derived from neighborhood-overlap similarity and communication-reliability information. The existence-side structure strength is intentionally kept much smaller than the spatial-side structure strength, so topology provides only a secondary regularization signal and does not dominate the core weight assignment.

The raw adaptive weights can fluctuate sharply over time, especially when packet delivery changes abruptly. To improve stability, the normalized weights are smoothed by an exponential moving average and then passed through a minimum-weight safeguard before final renormalization. This stabilization step is pragmatic rather than a novelty claim by itself; its purpose is to preserve the intended quality ordering while preventing unstable weight collapse in a time-varying communication setting.

4.9. Optional extension modules

The evaluated software also supports several exploratory extension modules outside the claimed core method. Their empirical effect is weaker, more coupled, or less stable than the evidence for the main three-factor backbone plus the retained branch-specific refinements, so they are kept out of the main method description and summarized only in Appendix B.

5. Experimental Setup

This section defines the evaluation protocol used to support the paper's claims. The experiments are organized around one main scenario and a small set of supporting studies. The main scenario is the dual-formation eight-sensor distributed GA-LMB tracking problem under tiered heterogeneous packet loss. Supporting studies are used to check whether the observed gains persist under ideal communication, how the method behaves as communication becomes more constrained, and whether several optional modules should remain outside the core method.

5.1. Evaluation hierarchy

The evaluation is intentionally hierarchical. The tiered-loss dual-formation scenario provides the confirmatory evidence for the proposed GA-LMB method family, while the factor ablation in the same scenario identifies which parts of the adaptive-weight design carry the result. Computational cost is measured on this main scenario as a deployment constraint rather than as a post hoc timing note. Two smaller supporting studies then test the boundaries of the claim: an ideal-communication comparison asks whether the refinement is useful without packet loss, and a communication-level sensitivity probe asks whether the same behavior persists as communication constraints become more severe. Additional exploratory variants are treated as appendix material rather than as competing headline methods.

5.2. Main scenario

The primary evaluation setting uses eight mobile sensors arranged as two four-sensor formations. The filtering mode is distributed local fusion, meaning that each node fuses only its own posterior and the posteriors available inside its communication neighborhood. The local multi-object filter is GA-LMB with loopy belief propagation (LBP) for approximate marginal data association (Williams and Lau, 2014).

The choice of the GA/KLA route is deliberate. GA fusion gives a conservative logarithmic-opinion-pool update under unknown cross-correlations, admits a stable canonical-form Gaussian fusion step for the LMB spatial densities, and naturally exposes the Bernoulli existence product rule in which branch-specific weighting can be applied. These properties make it a cleaner vehicle for studying adaptive weight allocation than a broad comparison across fusion families.

The common sensor-side configuration is:

- number of sensors: 8,
- filter mode: GA-LMB,
- data association: LBP,
- clutter rate: 3 for every sensor,
- detection probability: 0.9 for every sensor,
- measurement-noise scale: $q = 3$ for every sensor,
- sensor field of view enabled with half-angle 60° and range 60000.

The nominal detection probability, measurement-noise scale, and clutter rate are therefore homogeneous across sensors. The sensing variation exercised by the main scenario comes from the dual-formation geometry, finite fields of view, target visibility, and the resulting posterior covariance and existence differences; it should not be read as a sensitivity test over explicit p_D, measurement-noise, or clutter tiers.

The target side consists of ten objects arranged in three groups with staggered births. The three groups contain $3 + 3 + 4$ targets, respectively, and the targets move toward the central region with nominal speed 0.45. Births are staggered every eight time steps, and the total simulation length is 100 steps.

5.3. Communication model

The main scenario uses a tiered heterogeneous packet-loss setting. Communication is applied before distributed fusion through a measurement-delivery model with bandwidth limitation and link-level packet loss. The main configuration is:

- communication level: 2,

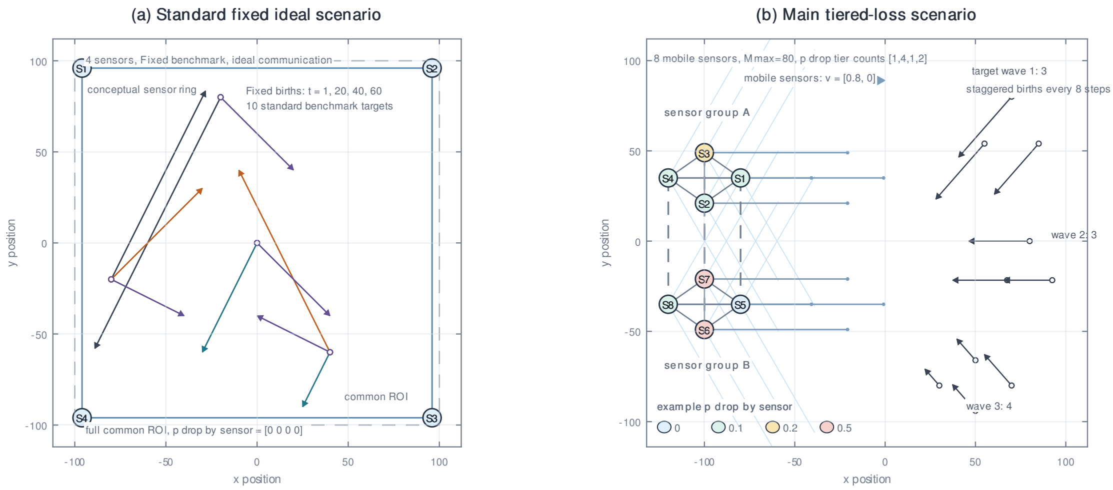

Figure 4: Schematic comparison between the standard fixed ideal benchmark and the main tiered heterogeneous packet-loss scenario. The ideal benchmark uses four full-ROI sensors with no packet loss and fixed target births; the main scenario uses the dual-formation eight-sensor geometry and staggered target waves with the bandwidth cap and persistent drop-tier heterogeneity used in the main experiments.

- global maximum delivered measurements per step: 80,
- packet-selection policy: weighted priority with random measurement selection,
- link model: fixed packet drop,
- target mean drop rate: $p_{\text{drop}} = 0.2$,
- tiered drop levels: $[0, 0.1, 0.2, 0.5]$,
- tiered level counts: $[1, 4, 1, 2]$.

This tiered design preserves the historical mean communication degradation while introducing persistent cross-node heterogeneity.

Figure 4 summarizes the contrast between the simpler standard fixed ideal benchmark and the main tiered-loss scenario. The standard ideal benchmark uses four full-ROI sensors, ideal communication, and fixed target births. The main scenario is the dual-formation eight-sensor case with staggered target waves, a bandwidth cap, and persistent drop-tier heterogeneity before distributed fusion.

5.4. Mainline ablation protocol

The core ablation study compares five consistently named arms:

1. Fixed Metropolis,
2. Covariance-only adaptive,
3. Covariance-link adaptive,
4. Three-factor adaptive backbone,
5. Balanced mode, which adds weak structure-aware modulation after covariance, link-quality, and existence-confidence weighting.

This five-arm path is a design diagnostic for the communication-aware backbone and the pre-refinement branch split. The manuscript results table keeps the visible rows focused on operating points by treating the three-factor backbone as the reliability model used by the branch-decoupled Balanced mode rather than as a separate deployment arm. The main confirmation additionally compares the FID-FIA baseline and the Cardinality-critical mode to show the branch-specific FID-FIA extension on the same scale. The Balanced mode is the position-sensitive operating mode. It uses the same parameterization across all trials: exponential smoothing with $\alpha = 0.7$, minimum weight 0.05, existence-confidence parameters $\lambda_{\min} = 0.85$ and $p_e = 2.0$, decoupled KLA strengths $(0.5, 0.15)$ for the spatial and existence branches, and weak structure-aware refinement strengths $(0.45, 0.08)$. The Cardinality-critical mode keeps the spatial side unchanged but applies FID-FIA information geometry only to the existence weights, with existence-branch smoothing and minimum-weight flooring disabled for that refinement. Exploratory extension modules are disabled in the main configuration.

5.5. Metrics and reporting

The experiments report three categories of metrics. The first category is network-level disagreement among sensors after distributed fusion. These primary metrics are OSPA consensus error, matched localization disagreement, and cardinality dispersion as defined in Section 3. The second category is conventional truth-referenced tracking accuracy. These secondary metrics are local E-OSPA, local RMSE, and local cardinality error. The third category is computational cost, measured as filter/fusion wall-clock runtime per trial, runtime per simulation step, and runtime relative to Fixed Metropolis. The OSPA/GOSPA family is used because it provides mathematically consistent finite-set distances for multi-object filtering and tracking evaluation (Schuhmacher et al., 2008; Rahmathullah et al., 2017; Ristic et al., 2011). The primary disagreement metrics are task-specific network-agreement measures built from these standard ingredients, not a replacement for standard truth-referenced tracking benchmarks, so the paper always interprets them together with the secondary local metrics and the computational-cost measurements rather than in isolation.

This hierarchy is deliberate. The proposed method is not intended as a generic single-sensor accuracy booster. It is designed as a distributed fusion-weight mechanism for reducing disagreement in the fused multi-sensor picture under heterogeneous communication. For that reason, the paper treats the network-level disagreement metrics as the main outcome, the local tracking metrics as a safeguard against hidden degradation, and computational cost as a first-class deployment-feasibility axis. The target is to improve cross-node agreement without causing a meaningful drop in conventional tracking quality or requiring impractical additional computation. Runtime is measured only around the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded so that the reported cost isolates the algorithmic overhead of adaptive weighting and branch-specific FID-FIA scoring.

Unless otherwise stated, reported scalar metrics are Monte Carlo means. The main fixed-versus-existence-refined confirmation, the five-arm factor ablation, the ideal-communication comparison, and the communication-level sensitivity study all use 50 deterministic paired trials with seeds 2–51. The main confirmation, ideal-communication comparison, and communication-level sensitivity study are paired by construction. The five-arm factor ablation remains primarily a design diagnostic, but it is now reported on the same 50-trial seed set for consistency.

6. Results and Discussion

This section reports the empirical evidence supporting the paper's main claims. The strongest evidence comes from the tiered heterogeneous packet-loss GA scenario introduced in Section 5, so that scenario is treated as the primary result. Ideal communication and communication-sensitivity experiments are used as supporting studies. Appendix-only results document exploratory variants without diluting the main GA/KLA method.

6.1. Main result in the tiered-drop GA scenario

The main claim is that communication-aware adaptive weighting substantially improves distributed consensus quality in the eight-sensor dual-formation GA-LMB scenario under tiered heterogeneous packet loss. Because the paper's design target is inter-sensor agreement, this subsection first reports the primary disagreement metrics and then checks the conventional local tracking metrics as a safeguard. The comparison separates three questions: whether dynamic reliability weighting is already beneficial, whether the communication-aware branch-decoupled design adds further value, and whether an existence-only information-geometric cue is useful when cardinality errors dominate. The two direct dynamic-weighting baselines answer the first question: the PD-weighted GA row uses detection-probability-style reliability weighting, and the target-wise FI-weighted GA row weights local Gaussian components by the trace of their information matrices.

Table 1 should therefore be read as an ordered pattern rather than as six isolated numeric rows. Both direct dynamic-weighting baselines improve over Fixed Metropolis, confirming that adaptive reliability information is useful under

Table 1
Main paired consensus comparison under tiered heterogeneous packet loss. Values are mean ± standard deviation over 50 deterministic paired trials with seeds 2–51. Lower values are better.

Arm	OSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 ± 0.220	2.326 ± 0.353	0.716 ± 0.224
PD-weighted GA	2.177 ± 0.161	1.995 ± 0.233	0.588 ± 0.170
FI-weighted GA	2.030 ± 0.128	1.870 ± 0.221	0.441 ± 0.127
FID-FIA baseline	1.818 ± 0.056	1.643 ± 0.109	0.123 ± 0.023
Balanced mode	1.779 ± 0.073	$\mathbf{1.522} \pm 0.157$	0.188 ± 0.031
Cardinality-critical mode	$\mathbf{1.678} \pm 0.050$	1.546 ± 0.158	$\mathbf{0.063} \pm 0.017$

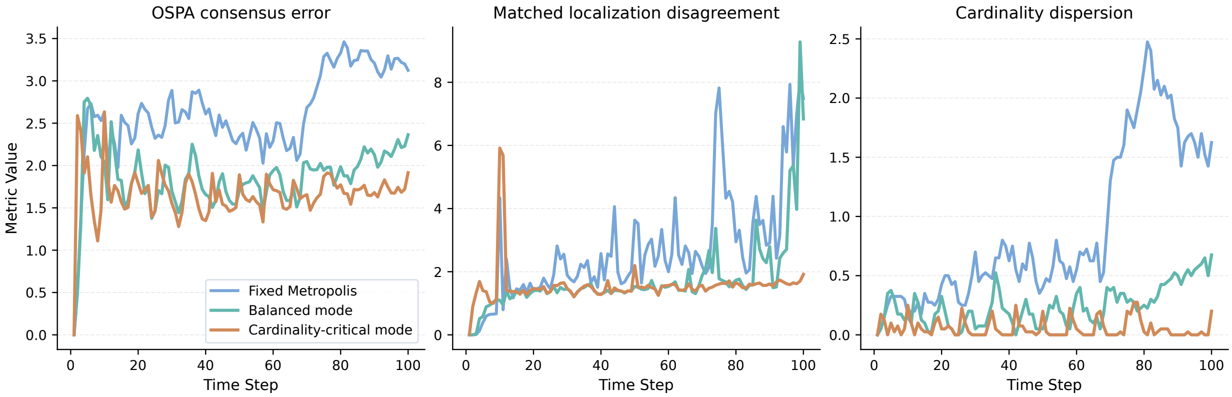

Figure 5: OSPA consensus error, matched localization disagreement, and cardinality dispersion over time for Fixed Metropolis, Balanced mode, and Cardinality-critical mode under tiered heterogeneous packet loss.

heterogeneous delivery. The communication-aware method family, however, provides the stronger consensus profile. The Balanced mode gives the best localization-oriented disagreement result, while the Cardinality-critical mode gives the lowest OSPA consensus error and cardinality dispersion: relative to Fixed Metropolis, it reduces OSPA consensus error from 2.469 to 1.678, matched localization disagreement from 2.326 to 1.546, and cardinality dispersion from 0.716 to 0.063. These correspond to reductions of 32.1%, 33.5%, and 91.2%, respectively. For the original method-family arms, the paired reductions relative to Fixed Metropolis are positive in all 50 trials for all three primary disagreement metrics, with exact sign-test $p = 1.8 \times 10^{-15}$. The FI-weighted GA result is therefore best interpreted as a useful external Fisher-information baseline, not as a substitute for the communication-aware branch-decoupled design. Figure 5 illustrates the same separation over time; both operating modes stay below the fixed baseline across most of the trajectory, with the Cardinality-critical mode most clearly separating on cardinality dispersion.

The primary-metric gain would be less convincing if it came at the cost of clear local tracking degradation. Table 2 provides this safeguard check. The FI-weighted GA baseline attains the lowest truth-referenced local RMSE, which is consistent with its target-wise spatial-information weighting, but it remains substantially weaker on local E-OSPA and local cardinality error. The Cardinality-critical mode shows the complementary behavior: it gives the best local E-OSPA and local cardinality error, while its RMSE is higher than the FI-weighted GA and Balanced rows but still slightly lower than the FID-FIA baseline. Thus the large cardinality-dispersion gain is not simply agreement around an incorrect target count. It is better understood as the effect of injecting information-geometric evidence into the existence branch while retaining the spatial-consensus advantages of the communication-aware adaptive backbone.

The two proposed operating-mode rows should therefore be read as recommended settings of the same method family rather than as an inconsistency in the proposed design. The Balanced mode is preferable when spatial RMSE stability is the dominant requirement. The Cardinality-critical mode is preferable when target-number agreement and truth-referenced cardinality error are the dominant risks. This usage-oriented distinction makes the tradeoff explicit instead of hiding it behind a single headline configuration.

Table 2
Conventional local tracking metrics in the main tiered heterogeneous packet-loss scenario. Values are mean ± standard deviation across the same 50 deterministic paired trials used in Table 1, after averaging each trial over sensors.

Arm	E-OSPA	RMSE	CardErr
Fixed Metropolis	2.863 ± 0.124	1.650 ± 0.078	1.455 ± 0.242
PD-weighted GA	2.736 ± 0.116	1.563 ± 0.061	1.255 ± 0.205
FI-weighted GA	2.496 ± 0.112	$\mathbf{1.548} \pm 0.050$	0.985 ± 0.175
FID-FIA baseline	2.185 ± 0.050	1.734 ± 0.094	0.388 ± 0.047
Balanced mode	2.335 ± 0.072	1.606 ± 0.050	0.579 ± 0.067
Cardinality-critical mode	$\mathbf{2.020} \pm 0.047$	1.721 ± 0.161	$\mathbf{0.224} \pm 0.029$

Table 3
Computational cost in the main tiered heterogeneous packet-loss scenario. Wall-clock time measures only the distributed LMB filtering/fusion call for each arm over 50 deterministic paired trials with seeds 2–51. The Fixed Metropolis, FID-FIA, Balanced, and Cardinality-critical rows come from the operating-mode runtime probe; the PD-weighted and FI-weighted GA rows come from the direct dynamic-weighting probe under the same scenario and seed set. Relative runtime is computed against the paired Fixed Metropolis denominator within the corresponding probe.

Arm	Runtime (s)	Runtime/step (s)	Relative runtime
Fixed Metropolis	52.123 ± 7.932	0.521	1.000×
PD-weighted GA	61.668 ± 5.376	0.617	1.233×
FI-weighted GA	62.810 ± 6.768	0.628	1.254×
FID-FIA baseline	147.674 ± 23.957	1.477	2.833×
Balanced mode	56.378 ± 9.626	0.564	1.082×
Cardinality-critical mode	155.913 ± 18.220	1.559	2.991×

Computational cost is the third evaluation axis for this method family. Table 3 reports 50-trial runtime measurements in the same 100-step tiered-drop scenario and covers all arms in the main comparison. The measurement includes only the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded. The main cost distinction is clear. The Balanced mode remains close to the fixed-weight baseline, requiring about 1.08× the Fixed Metropolis runtime. The PD-weighted and FI-weighted GA baselines are also moderate-cost dynamic-weighting alternatives at about 1.23×–1.25× fixed runtime. By contrast, the FID-FIA baseline and the Cardinality-critical mode are both near the three-times-cost regime. The runtime evidence therefore identifies FID-FIA scoring, not the basic communication-aware weighting backbone or the direct PD/FI dynamic-weight baselines, as the dominant source of additional cost.

The cost results sharpen the operating-mode recommendation. The Balanced mode is the low-overhead default when spatial consensus, RMSE stability, runtime, or energy budget is the limiting requirement. The Cardinality-critical mode should be selected when target-number agreement and suppression of missed or false support justify roughly three times the fixed-weight filtering/fusion time.

6.2. Factor ablation of the main adaptive design

The most important diagnostic ablation is the factor-by-factor path from Fixed Metropolis to the Balanced mode. Table 4 summarizes this 50-trial path and adds the Cardinality-critical mode as the last row. To keep the table focused on the main operating points, the three-factor adaptive backbone is not listed as a separate row; it remains the reliability model used by the branch-decoupled Balanced mode. The last row is the branch-specific existence refinement that converts the Balanced mode into the cardinality-sensitive mode.

Several conclusions follow from Table 4. First, covariance weighting is necessary but not sufficient. It improves over fixed topology weights, confirming that posterior concentration is a useful first-order proxy for local posterior quality, but it still leaves a substantial gap to the branch-decoupled operating modes.

Second, realized link quality is the dominant communication-aware factor under heterogeneous packet loss. Adding it after covariance produces the largest incremental change in the backbone path, especially for cardinality dispersion,

Table 4
Network disagreement metrics for the main factor ablation under tiered heterogeneous packet loss. Values are mean ± standard deviation over 50 deterministic paired trials. The rows show the progression from fixed weighting to the covariance-link backbone and then to the Balanced operating mode, which includes the three-factor adaptive backbone and weak structure-aware branch decoupling. The final row applies the FID-FIA cue only to the existence-weight path to obtain the cardinality-critical operating mode.

Arm	OSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.469 ± 0.220	2.326 ± 0.353	0.716 ± 0.224
Covariance-only adaptive	2.091 ± 0.145	1.951 ± 0.253	0.466 ± 0.134
Covariance-link adaptive	1.788 ± 0.071	1.525 ± 0.156	0.188 ± 0.031
Balanced mode	1.779 ± 0.073	$\mathbf{1.522} \pm 0.157$	0.188 ± 0.031
Cardinality-critical mode	$\mathbf{1.678} \pm 0.050$	1.546 ± 0.158	$\mathbf{0.063} \pm 0.017$

which is approximately halved. This is the key evidence that the method is not merely reweighting sharper posteriors; it is also responding to whether those posteriors are reliably delivered to neighboring nodes.

Third, existence confidence remains part of the adaptive backbone, but its effect is not isolated as a separate operating row in Table 4. The reason is methodological rather than negative: existence confidence is used together with branch decoupling in the Balanced mode, where the reliability model is applied differently to the spatial and Bernoulli existence paths. This is consistent with the design rationale: covariance and link quality do not directly express how decisive a local Bernoulli component is about target existence.

Finally, the Balanced mode provides the pre-FID-FIA branch-decoupled operating point. The final row changes the operating point by applying the FID-FIA cue only to the existence branch, reducing cardinality dispersion from the approximately 0.188 level of the Balanced/backbone setting to 0.063 while also lowering OSPA consensus error. This is why the proposed method is presented as a configurable branch-decoupled family: the Balanced mode is the spatial/position-oriented setting, and the Cardinality-critical mode is the cardinality-oriented setting.

6.3. Ideal-communication supporting evidence

An obvious question is whether the weak structure-aware decoupled refinement only helps because it compensates for packet loss. The ideal-communication experiment addresses this point by setting the communication level to zero and comparing Ordinary GA, the Balanced mode, the FID-FIA baseline, and the Cardinality-critical mode under otherwise matched conditions. Table 5 reports this comparison over the same 50 deterministic seed set used in the main confirmation.

Table 5
Ideal-communication supporting comparison over 50 deterministic paired trials with seeds 2–51. Adaptive rows report the mean value and relative change versus ordinary GA; lower values are better, so negative percentages indicate improvements.

	Network disagreement			Local safeguards	
Arm	OSPA err.	Loc. disag.	Card. disp.	E-OSPA	RMSE
Ordinary GA	1.634	1.381	0.090	1.908	1.442
Balanced mode	**1.427 (-12.7%)**	**1.202 (-13.0%)**	0.071 (-20.9%)	1.839 (-3.6%)	**1.375 (-4.7%)**
FID-FIA baseline	1.534 (-6.1%)	1.333 (-3.5%)	0.057 (-36.3%)	1.881 (-1.4%)	1.500 (+4.0%)
Cardinality-critical mode	1.433 (-12.3%)	1.303 (-5.7%)	**0.049 (-46.0%)**	**1.766 (-7.4%)**	1.470 (+2.0%)

The ideal-communication result mirrors the branch specialization observed in the tiered-drop experiment. Information-geometric weighting is especially effective for cardinality dispersion, whereas the Balanced mode remains the localization-oriented alternative and gives the lowest OSPA consensus error, matched localization disagreement, and local RMSE in this benign setting. The Cardinality-critical mode gives the lowest cardinality dispersion and local E-OSPA without forcing a single FID-FIA weight onto the whole posterior. This supporting study therefore serves two purposes: it shows that the refinement is not merely repairing packet loss, and it shows that the consensus gain is not purchased by a collapse in local tracking quality.

Table 6
Communication-level sensitivity in the eight-sensor dual-formation GA-LMB scenario. Each level uses 50 deterministic paired trials. Values are Fixed Metropolis / Balanced mode / Cardinality-critical mode, and lower values are better.

Level	Additional constraint	OSPA err.	Loc. disag.	Card. disp.
0	none	1.628/1.423/1.432	1.378/1.201/1.303	0.087/0.069/0.048
1	bandwidth cap	1.701/1.482/1.473	1.447/1.242/1.344	0.126/0.085/0.049
2	tiered link loss	2.383/1.760/1.669	2.263/1.536/1.571	0.650/0.180/0.066
3	node outage	2.656/1.808/1.689	2.701/1.570/1.574	0.880/0.201/0.066

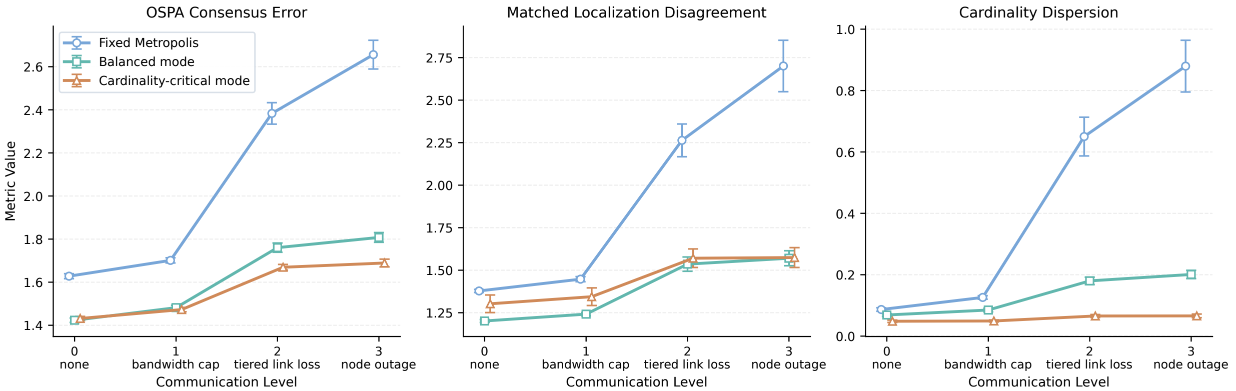

Figure 6: Communication-level sensitivity for Fixed Metropolis, Balanced mode, and Cardinality-critical mode. Markers show 50-trial means and error bars show 95% confidence intervals.

6.4. Communication sensitivity and robustness

A communication-level sensitivity study was added to test whether the two operating modes remain useful outside the single tiered-drop setting. The study uses the same eight-sensor dual-formation GA-LMB scenario and compares Fixed Metropolis, Balanced mode, and Cardinality-critical mode over 50 deterministic paired trials at each communication level. Level 0 is ideal communication, level 1 adds only the global bandwidth cap, level 2 adds the tiered fixed link-loss model used in the main experiment, and level 3 further adds random node outages. Table 6 reports the resulting network-level disagreement metrics, and Figure 6 visualizes the same means with 95% confidence intervals. This table is a communication-sensitivity check; it does not test explicit sensor-side p_D or measurement-noise tiers.

This study is not used to replace the main tiered-drop comparison, but it closes the previous gap in the communication-sensitivity evidence. Both proposed operating modes improve on Fixed Metropolis at every communication level. The important pattern is the gradient of the effect: the gain is modest under ideal or bandwidth-only communication, but it becomes much larger once tiered link loss or node outages make realized delivery quality heterogeneous. The Balanced mode gives the lowest matched localization disagreement at every level, while the Cardinality-critical mode gives the lowest cardinality dispersion at every level and the lowest OSPA consensus error once communication constraints are nontrivial. The local safeguard metrics move in the same direction: local E-OSPA and local cardinality error are lowest for the Cardinality-critical mode, while local RMSE remains strongest for the Balanced mode. This supports the interpretation that communication-aware weighting is most valuable when communication quality varies across nodes, and that the two operating modes expose a meaningful spatial-versus-cardinality tradeoff rather than a single brittle tuning.

7. Conclusion

This paper studies distributed KLA-based (geometric-average) LMB fusion for multi-sensor multi-object tracking under communication constraints. The starting point is that Fixed Metropolis weighting is too rigid for networks in which posterior quality and effective communication quality vary across nodes. In the present peer-to-peer setting, the most useful question is therefore not whether geometric-average fusion should be abandoned, but how its weights should be allocated so that they reflect both local posterior quality and realized network conditions.

The experiments support a focused conclusion. The most effective adaptive design is built on three factors: covariance quality, realized link quality, and existence confidence. Covariance captures posterior concentration, link quality captures how reliably a node's information is actually delivered, and existence confidence captures a missing dimension that is not expressed by covariance alone, namely how decisively a local Bernoulli posterior supports target existence. On top of this three-factor backbone, branch decoupling lets the spatial path retain the communication-aware reliability model while the existence path can receive the FID-FIA cue that is empirically strongest for cardinality. These refinements act as branch-specific corrections rather than as topology-dominated or information-geometry-only replacements for the core adaptive weighting rule.

In the dual-formation eight-sensor scenario under tiered heterogeneous packet loss, this design yields clear improvements in distributed consensus quality. The headline result is not simply a lower average score on one metric: the Cardinality-critical mode improves OSPA consensus error, matched localization disagreement, and cardinality dispersion together, with the largest effect on the target-number disagreement that fixed topology weights struggle to control. The local tracking safeguards support the same interpretation. Mean local E-OSPA and local cardinality error both decrease substantially relative to Fixed Metropolis, while the RMSE tradeoff remains moderate and comparable to the information-geometric baseline.

The runtime supplement adds the deployment side of this conclusion. The Balanced mode stays close to fixed-weight cost at about 1.08× fixed runtime, and the direct PD-weighted and FI-weighted GA baselines remain moderate-cost alternatives at about 1.23×–1.25×. The FID-FIA baseline and Cardinality-critical mode instead require roughly three times the fixed-weight filtering/fusion time. The ablation study explains this tradeoff: covariance and link quality form the essential communication-aware backbone, existence confidence adds target-existence decisiveness, and branch-specific FID-FIA refinement separates the low-overhead position-sensitive setting from the higher-cost cardinality-sensitive setting.

The supporting studies help clarify what this result does and does not mean. Under ideal communication, both operating modes still improve network disagreement metrics relative to ordinary distributed GA, so the gain is not only a packet-loss compensation effect. Across the communication-level sensitivity probe, the advantage is largest when tiered link loss or node outage makes realized delivery heterogeneous. By contrast, the present experiments keep nominal detection probability, clutter rate, and measurement-noise scale homogeneous across sensors, so richer sensor-parameter heterogeneity remains a future-work direction rather than a claimed generalization result. Exploratory variants outside the branch-decoupled communication-aware design show weaker or mixed evidence and are therefore kept outside the main claimed method.

The present work therefore supports a focused conclusion: communication-aware adaptive fusion for distributed KLA-based LMB tracking benefits most from combining posterior concentration, realized communication reliability, and existence decisiveness, then applying branch-specific refinements that reserve information-geometry scoring for existence/cardinality fusion when the deployment is cardinality-critical. In deployment terms, the Balanced mode should be selected when RMSE, spatial stability, or computation budget is the limiting requirement, whereas the Cardinality-critical mode should be selected when target-number agreement is the limiting requirement and roughly three times the fixed-weight filtering/fusion runtime is acceptable. Future work should strengthen this conclusion through larger Monte Carlo studies, stronger external baselines, richer heterogeneous sensing scenarios, runtime optimization of the FID-FIA branch, and more principled dynamic structure models. A particularly useful extension is asynchronous or multi-rate distributed LMB fusion, where availability terms could be replaced by time-aligned posterior propagation and continuous-time Fisher-information decay, and where target-wise information weights could be revisited once label association across neighboring LMB posteriors is sufficiently reliable.

A. Simulation Setup

This appendix summarizes the simulation settings used in the experiments. The purpose is reproducibility rather than narrative emphasis. The main text explains why the experiments were chosen; the present appendix records the concrete scenario geometry, target initialization, communication model, and supporting ideal-communication variant.

A.1. Main scenario overview

The primary experiment uses an eight-sensor distributed formation-tracking scenario composed of two four-sensor formations. Each node runs a local GA-LMB filter with loopy belief propagation (LBP) for approximate marginal data association (Williams and Lau, 2014), and distributed fusion is performed only over the local communication neighborhood.

The common filter-side settings are:

- number of sensors: 8,
- local filter mode: GA-LMB,
- data association: LBP,
- simulation length: 100 time steps,
- sensor communication range: 150,
- fixed-weight baseline fusion rule: Metropolis-type consensus weighting (Xiao et al., 2005),
- clutter rate at every sensor: 3,
- detection probability at every sensor: 0.9,
- measurement-noise scale at every sensor: $q = 3$,
- field of view enabled with half-angle 60° and range 60000.

These are homogeneous nominal sensor settings. Geometry and field-of-view gating create target- and time-dependent visibility differences, but no explicit per-sensor p_D, noise, or clutter tiers are used in the main reported experiment.

A.2. Sensor geometry and neighborhood graph

The eight sensors are divided into two groups of four. Both groups follow constant-velocity platform trajectories with zero process noise and move from left to right with nominal velocity $[0.8, 0]^\mathsf{T}$. The two formation centers are $[-80, 35]^\mathsf{T}$ and $[-80, -35]^\mathsf{T}$, respectively. Each group uses a four-agent leader-follower pattern with one lead sensor and three offset followers, with spacing 20.

The local communication graph is fixed. Each formation is fully connected internally, and the two formations are connected by one-to-one cross links: $1 \leftrightarrow 5$, $2 \leftrightarrow 6$, $3 \leftrightarrow 7$, and $4 \leftrightarrow 8$. The corresponding neighborhood sets are therefore $\{1, 2, 3, 4, 5\}$, $\{1, 2, 3, 4, 6\}$, $\{1, 2, 3, 4, 7\}$, $\{1, 2, 3, 4, 8\}$, $\{1, 5, 6, 7, 8\}$, $\{2, 5, 6, 7, 8\}$, $\{3, 5, 6, 7, 8\}$, and $\{4, 5, 6, 7, 8\}$.

A.3. Target initialization

The target side contains ten objects divided into three groups with staggered births. The birth interval is eight time steps, the birth start time is one, and the target lifetime is 100 time steps. The three groups are: three targets in a Triangle formation centered at $[70, 80]^\mathsf{T}$ with spacing 30, three targets in a Triangle formation centered at $[80, 0]^\mathsf{T}$ with spacing 25, and four targets in a four-agent leader-follower pattern centered at $[70, -80]^\mathsf{T}$ with spacing 20. All three groups move toward the central region with speed 0.45.

A.4. Communication model

Communication degradation is applied before distributed fusion through a model that combines a global per-step measurement budget with optional link-level packet loss and optional node outages. The communication levels are:

- level 0: no communication restriction,
- level 1: global measurement budget only,
- level 2: level 1 plus link-level packet loss,
- level 3: level 2 plus node outages.

The main paper scenario uses level 2 with global maximum delivered measurements per step 80, fixed link loss, nominal mean packet-drop rate $p_{\text{drop}} = 0.2$, and tiered heterogeneous packet-loss configuration

$$p_{\text{drop}} \in [0,\ 0.1,\ 0.2,\ 0.5], \qquad \text{counts} = [1,\ 4,\ 1,\ 2]. \tag{28}$$

This tiered configuration preserves the historical mean packet-drop rate while creating persistent node-to-node communication heterogeneity.

A.5. Ideal-communication supporting variant

The supporting ideal-communication comparison uses the same sensor geometry, target initialization, and local filtering configuration as the main scenario, but removes communication degradation. The relevant settings are communication level 0, effectively unlimited delivery budget, and $p_{\text{drop}} = 0$ for all sensors. The same deterministic seed range is used for Ordinary GA, the Balanced mode, the FID-FIA baseline, and the Cardinality-critical mode.

A.6. Main adaptive configurations

The Balanced mode used in the main tiered-drop comparison is:

- `emaAlpha` $= 0.7$,
- `minWeight` $= 0.05$,
- covariance, link-quality, and existence-confidence terms enabled,
- `existenceConfidenceMinScore` $= 0.85$,
- `existenceConfidencePower` $= 2.0$,
- decoupled KLA enabled,
- weak structure-aware refinement enabled,
- FID-FIA refinement disabled,
- posterior-structure-consistency, NIS, and history disabled in the main configuration,
- spatial decoupling and structure strengths $(0.5, 0.45)$,
- existence decoupling and structure strengths $(0.15, 0.08)$.

The Cardinality-critical mode uses the same settings and additionally enables FID-FIA refinement only for the existence branch.

A.7. Monte Carlo seed control

The main fixed-versus-existence-refined confirmation, the supporting ideal-communication comparison, the factor ablation, and the communication-level sensitivity study all use 50 deterministic paired trials with seeds 2–51. The Cardinality-critical mode row reported with the ablation uses the same deterministic seed range and scenario protocol. For every trial, the same generated truth, measurements, communication realization, and delivered measurement sets are used for all compared fusion arms within that trial.

B. Additional Attempted Modules and Results

This appendix records secondary routes and modules that were evaluated but not retained as part of the main claimed method. The purpose is to document that the design space was explored beyond the main GA/KLA method family and to bound the empirical value of alternatives that are not promoted into the paper body.

B.1. AA-based secondary route

The paper body focuses on GA/KLA fusion because the method is designed around conservative logarithmic pooling and branch-specific GA-LMB fusion weights. A secondary arithmetic-average experiment was nevertheless run in the same eight-sensor communication-constrained setting with a three-wave target arrangement. The result is positive but is kept here because it is a different fusion route and is not based on the final existence-refined GA configuration.

Table 7
Appendix-only AA-based secondary route.

Arm	OSPA	RMSE	Cardinality
fixed AA	4.349	19.098	0.421
adaptive AA	3.811	16.472	0.307

This result suggests that communication-aware weighting may also be useful outside GA fusion, but it is not used as a main claim. It is retained only as appendix evidence because the theoretical development, operating-mode story, and strongest experiments in this paper are all built around distributed GA/KLA-based LMB fusion.

B.2. NIS-based consistency weighting

This appendix-level experiment studies whether innovation-based consistency information should enter the adaptive weight model. NIS is a standard filter-diagnostic statistic in tracking and navigation, and recent work also uses related distributional checks for estimator auto-tuning (Bar-Shalom, Li and Kirubarajan, 2001; Chen, Biggie, Ahmed, Julier and Heckman, 2023). The relevant comparison here is `w/o NIS` → `robust NIS` → `plain NIS`. The NIS term is treated as a consistency penalty rather than a monotonic quality reward because innovation consistency is structurally coupled with covariance-based posterior quality.

Table 8
Network disagreement metrics for the NIS-based consistency comparison.

Arm	OSPA	Loc. disag.	Cardinality
w/o NIS	1.792	1.556	0.198
robust NIS	1.804	1.552	0.203
plain NIS	1.897	1.739	0.237

Plain NIS is clearly harmful in the present setting. Robust NIS remains substantially better than plain NIS, but across 50 trials it is still slightly worse than the no-NIS baseline in OSPA and cardinality dispersion, with only a small localization-disagreement improvement.

B.3. Freshness weighting

This experiment tests whether a recency-oriented score improves fusion beyond the robust-NIS baseline. The intended comparison is robust NIS baseline versus robust NIS plus freshness. In the 50-trial comparison, the result is effectively flat: OSPA remains at 1.804, matched localization disagreement changes from 1.552 to 1.551, and cardinality dispersion remains at 0.203.

B.4. History weighting

This experiment evaluates whether a temporal-stability score improves fusion quality. In the exploratory comparison, OSPA changes from 1.811 to 1.814, localization disagreement changes from 3.173 to 3.158, and cardinality dispersion changes from 0.214 to 0.215. This effect is too small and too mixed to justify a central role in the main text.

B.5. Cardinality-consensus weighting

An exploratory direct cardinality-consensus score was also tested. Relative to the covariance-and-link baseline, the pilot changed OSPA from 1.909508 to 1.954855, RMSE from 1.621662 to 1.791287, and cardinality dispersion from 0.242500 to 0.285000. Because this evidence is negative and only a pilot, it is retained only as appendix context.

B.6. Association ambiguity and posterior-structure consistency

Two additional extensions were also evaluated. The association-ambiguity add-on changes the baseline OSPA, RMSE, and cardinality dispersion from (1.874840, 1.779820, 0.244500) to (1.876368, 1.769102, 0.245500), which is too mixed to justify promotion into the main text. The stronger posterior-structure-consistency extension remains tied with the retained weak static structure prior at (1.862244, 1.749608, 0.244250), so it is best treated as a future extension candidate rather than a present claim.

Taken together, these appendix results strengthen the main paper rather than weaken it. They show that the design space was explored beyond the final chosen method, and they justify why the present paper is intentionally centered on covariance quality, realized link quality, existence confidence, branch decoupling, and the existence-branch FID-FIA cue.

C. Theoretical Interpretation of Adaptive Branch-Decoupled Fusion

This appendix records the theoretical interpretation of the adaptive branch-decoupled fusion rule used in the main paper. The purpose is narrower than a full new optimality theory. We focus on the exact statements that are supported by the proposed method and the standard KLA/LMB background (Battistelli and Chisci, 2014; Hlinka et al., 2014; Üney et al., 2013; Bandyopadhyay and Chung, 2018; Vo et al., 2014; Reuter et al., 2014; Vo et al., 2019). Throughout this appendix, we assume the same label alignment and moment-compatible Gaussian projection used by the merger.

C.1. Entropy-Regularized Interpretation of the Weight Backbone

Fix a fusion node s at time k with active neighborhood $\mathcal{A}_{k,s}$. The adaptive backbone constructs the positive score

$$s_{0,j} = q_{\mathrm{cov},k}^{(j)} q_{\mathrm{link},k}^{(j)} q_{\mathrm{exist},k}^{(j)}, \qquad j \in \mathcal{A}_{k,s}, \tag{29}$$

where q_{cov}, q_{link}, and q_{exist} denote the covariance-quality, realized link-quality, and existence-confidence terms, respectively. Define

$$u_{0,j} = \log q_{\mathrm{cov},k}^{(j)} + \log q_{\mathrm{link},k}^{(j)} + \log q_{\mathrm{exist},k}^{(j)}. \tag{30}$$

Let

$$\Delta_{k,s} = \left\{ w \in \mathbb{R}^{|\mathcal{A}_{k,s}|} : w_j \geq 0, \ \sum_j w_j = 1 \right\} \tag{31}$$

be the active-neighbor simplex.

Proposition C1. For any temperature $\tau > 0$, the optimization problem

$$\max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j u_{0,j} + \tau H(w) \right\}, \qquad H(w) = -\sum_j w_j \log w_j, \tag{32}$$

has the unique optimizer

$$w_j^\star = \frac{\exp(u_{0,j}/\tau)}{\sum_u \exp(u_{0,u}/\tau)} = \frac{s_{0,j}^{1/\tau}}{\sum_u s_{0,u}^{1/\tau}}. \tag{33}$$

In particular, when $\tau = 1$, the implemented normalized product rule is recovered:

$$w_j^\star = \frac{s_{0,j}}{\sum_u s_{0,u}}. \tag{34}$$

Thus the multiplicative backbone is exactly an entropy-regularized simplex allocation in log-utility space, not merely an empirical product of factors.

Isotropic interpretation of the covariance term. If the local Gaussian components satisfy

$$T_{k,i}^{(j)} = \sigma_j^2 I_d \tag{35}$$

for all labels i at sensor j, then the implemented covariance score

$$q_{\mathrm{cov},k}^{(j)} = \frac{1}{\epsilon + d\sigma_j^2} \tag{36}$$

induces exactly the same ordering as smaller σ_j^2, smaller $\log\det T_{k,i}^{(j)}$, larger $\log\det(T_{k,i}^{(j)})^{-1}$, and larger $\mathrm{tr}((T_{k,i}^{(j)})^{-1})$. Hence, in the isotropic limit, the retained trace-based score agrees with standard information criteria. In near-isotropic regimes, the disagreement with log-determinant criteria enters only through higher-order anisotropy terms.

C.2. Exact Branch Structure of the Implemented GA-LMB Merger

The implemented merger uses separate spatial and existence weights. This separation is not purely heuristic; it is supported by the algebraic structure of the GA-LMB update.

Spatial branch. For a fixed Bernoulli component i, the Gaussian canonical fusion step is

$$K_i = \sum_j w_{x,j} T_{i,j}^{-1}, \tag{37}$$

$$h_i = \sum_j w_{x,j} T_{i,j}^{-1} v_{i,j}, \tag{38}$$

with

$$\Sigma_i = K_i^{-1}, \qquad \mu_i = \Sigma_i h_i. \tag{39}$$

Proposition C2. Conditional on the spatial weights w_x, the Gaussian density $\mathcal{N}(\mu_i, \Sigma_i)$ is the unique minimizer of

$$\min_{q\in\mathcal{G}} \sum_j w_{x,j} D_{\mathrm{KL}}\left(q \,\|\, \mathcal{N}(v_{i,j}, T_{i,j})\right), \tag{40}$$

where $\mathcal{G}$ is the Gaussian family. Hence the implemented spatial update is the exact Gaussian KLA barycenter.

Existence branch. The same merger computes

$$r_i = \frac{\eta_i(w_x)\prod_j r_{i,j}^{w_{r,j}}}{\eta_i(w_x)\prod_j r_{i,j}^{w_{r,j}} + \prod_j (1-r_{i,j})^{w_{r,j}}}, \tag{41}$$

where $\eta_i(w_x)$ is the spatial-overlap normalization term induced by the Gaussian branch.

Proposition C3. The fused existence probability satisfies the exact log-odds identity

$$\log\frac{r_i}{1-r_i} = \log\eta_i(w_x) + \sum_j w_{r,j}\log\frac{r_{i,j}}{1-r_{i,j}}. \tag{42}$$

Therefore, for fixed spatial weights, the existence branch is an exact weighted logit pool with an additive spatial-overlap offset.

C.3. From Bernoulli-RFS Decomposition to an Aligned-LMB Surrogate

The key theoretical bridge is that Bernoulli-RFS divergence already separates into existence and spatial terms.

Proposition C4. Let $\beta = (r, p)$ and $\beta_j = (r_j, p_j)$ be Bernoulli RFS densities. Then

$$D_{\mathrm{KL}}(\beta \,\|\, \beta_j) = d_{\mathrm{Ber}}(r \,\|\, r_j) + r\, D_{\mathrm{KL}}(p \,\|\, p_j), \tag{43}$$

where

$$d_{\mathrm{Ber}}(r \,\|\, r_j) = (1-r)\log\frac{1-r}{1-r_j} + r\log\frac{r}{r_j}. \tag{44}$$

This decomposition shows that spatial mismatch is naturally weighted by existence level.

Proposition C5. For a fixed Bernoulli component i, consider the decoupled surrogate

$$\mathcal{J}_i(r, p; w_x, w_r) = \sum_j w_{r,j}\, d_{\mathrm{Ber}}(r \,\|\, r_{i,j}) + r \sum_j w_{x,j}\, D_{\mathrm{KL}}(p \,\|\, p_{i,j}). \tag{45}$$

Its minimizer satisfies

$$p_i^\star(x) = \frac{1}{\eta_i(w_x)} \prod_j p_{i,j}(x)^{w_{x,j}}, \tag{46}$$

$$\log\frac{r_i^\star}{1-r_i^\star} = \log\eta_i(w_x) + \sum_j w_{r,j}\log\frac{r_{i,j}}{1-r_{i,j}}. \tag{47}$$

Hence the implemented spatial and existence update rules are the exact optimizer of a per-Bernoulli decoupled surrogate objective.

If local posteriors share an aligned label set and are represented in product LMB form, then the reverse-KL divergence is additive across labels. Under this same alignment assumption, the per-Bernoulli surrogate lifts directly to an aligned-LMB surrogate objective that decomposes labelwise.

Scope boundary. This appendix does not claim that the branch-decoupled rule is the unique optimizer of a fully unconstrained LMB fusion objective. The exact derivation above still relies on the same two architectural assumptions used by the merger: aligned labels and moment-compatible spatial projection.

C.4. Branch-Decoupled Weight Allocation and Structure-Aware Regularization

The branch-specific scores are

$$s_{x,j} = \left(q_{\mathrm{cov},j}\right)^{\alpha_x} \left(q_{\mathrm{link},j}\right)^{\beta_x}, \tag{48}$$

$$s_{r,j} = \left(q_{\mathrm{link},j}\right)^{\beta_r} \left(q_{\mathrm{exist},j}\right)^{\alpha_r}, \tag{49}$$

with geometric interpolation

$$\tilde{s}_{x,j} = s_{0,j}^{1-\eta_x} s_{x,j}^{\eta_x}, \tag{50}$$

$$\tilde{s}_{r,j} = s_{0,j}^{1-\eta_r} s_{r,j}^{\eta_r}. \tag{51}$$

This is exactly a convex interpolation in log-utility space. Define $\hat{u}_{x,j} = \log\tilde{s}_{x,j}$ and $\hat{u}_{r,j} = \log\tilde{s}_{r,j}$ for the branch utilities before graph and FID-FIA refinement.

The FID-FIA existence refinement adds one more nonnegative factor only to the existence branch. Let a_j denote the Fisher-information-distance accumulation score computed from existence-weighted local target pairs, and define the bounded normalized score

$$q_{\mathrm{fid},j} = \rho_{\min} + (1-\rho_{\min})\left(\frac{a_j}{\epsilon + \max_{u\in\mathcal{A}_{k,s}} a_u}\right)^{p_f}. \tag{52}$$

The refined existence score is

$$\tilde{s}_{r,j}^{\mathrm{fid}} = \tilde{s}_{r,j}\, q_{\mathrm{fid},j}^{\gamma_f}. \tag{53}$$

In log-utility form, this is simply

$$\hat{u}_{r,j}^{\mathrm{fid}} = \hat{u}_{r,j} + \gamma_f \log q_{\mathrm{fid},j}, \tag{54}$$

while the spatial utility $\hat{u}_{x,j}$ is unchanged. The log-utility statements below are exact for $\rho_{\min} > 0$; the main experiment setting $\rho_{\min} = 0$ is the boundary limit that allows the existence branch to place zero instantaneous weight on sensors whose FID-FIA accumulation is zero.

After decoupling, the structure-aware refinement multiplies the spatial score by $\xi_{x,j}^{\gamma_x}$ and the existence score, or the FID-FIA-refined existence score when enabled, by $\xi_{r,j}^{\gamma_r}$. This is equivalent to adding $\gamma_x \log \xi_{x,j}$ and $\gamma_r \log \xi_{r,j}$ to the corresponding branch utilities.

Proposition C6. Let p_x and p_r be normalized priors satisfying

$$p_{x,j} \propto \xi_{x,j}^{\gamma_x/\tau_x}, \tag{55}$$

$$p_{r,j} \propto \xi_{r,j}^{\gamma_r/\tau_r}. \tag{56}$$

Then the branch-wise optimization problems are equivalently written as

$$w_x^\star = \arg \max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{x,j} - \tau_x D_{\mathrm{KL}}(w \,\|\, p_x) \right\}, \tag{57}$$

$$w_r^\star = \arg \max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{r,j} - \tau_r D_{\mathrm{KL}}(w \,\|\, p_r) \right\}. \tag{58}$$

Thus the weak structure-aware term is precisely a KL regularization toward a graph-induced prior on the simplex.

Corollary C6a. With the FID-FIA existence refinement, the same branch-wise entropy-regularized program remains valid after replacing the existence utility $\hat{u}_{r,j}$ by $\hat{u}_{r,j}^{\mathrm{fid}}$. Equivalently,

$$w_{r,j}^\star \propto \tilde{s}_{r,j} \xi_{r,j}^{\gamma_r} q_{\mathrm{fid},j}^{\gamma_f}, \qquad w_{x,j}^\star \propto \tilde{s}_{x,j} \xi_{x,j}^{\gamma_x}. \tag{59}$$

Thus the FID-FIA term is theoretically just another existence-branch log-utility, not a change to the spatial KLA barycenter.

Proposition C7. The statement below applies to the unrefined existence utility $\hat{u}_{r,j}$ and, after the substitution in Corollary C6a, to the FID-FIA-refined utility $\hat{u}_{r,j}^{\mathrm{fid}}$. Define the branch utilities

$$U_x(w_x) = \sum_j w_{x,j} \hat{u}_{x,j} + \tau_x H(w_x), \tag{60}$$

$$U_r(w_r) = \sum_j w_{r,j} \hat{u}_{r,j} + \tau_r H(w_r). \tag{61}$$

Then the branch-decoupled adaptive weights are the exact optimizer of the block-separable surrogate program

$$\max_{w_x, w_r \in \Delta_{k,s}} \left\{ U_x(w_x) + U_r(w_r) \right\}. \tag{62}$$

Moreover, together with the aligned-LMB surrogate state-update block above, the merger and the adaptive weight-allocation rule are consistent with a single block-relaxed objective whose two blocks reproduce the state-update and weight-update steps.

Scalar-FID-FIA as a constrained special case. Using a single FID-FIA scalar weight vector for the whole fused posterior can be viewed as imposing the additional constraint $w_x = w_r = w_{\mathrm{fid}}$ and choosing both branch utilities to be dominated by $\log q_{\mathrm{fid},j}$. The proposed refinement relaxes this constraint: w_x remains governed by the communication-aware spatial utility, while w_r receives the FID-FIA term. This provides a theoretical explanation for the observed empirical tradeoff. A single FID-FIA weight can improve cardinality by moving the existence logit pool toward sensors with high pairwise Fisher separability, but the same move also perturbs the Gaussian precision barycenter. Branch decoupling keeps the information-geometric signal where it is most useful, in the Bernoulli existence pool, without forcing the spatial barycenter to use the same weights.

Proposition C7a. Let p_x and p_r be strictly positive priors on the simplex, and let Q_x and Q_r be differentiable latent branch-quality functionals. The entropic proximal linearizations

$$w_x^+ = \arg\max_{w\in\Delta_{k,s}} \left\{\langle \nabla Q_x(p_x),\, w - p_x\rangle - \tau_x D_{\mathrm{KL}}(w \,\|\, p_x)\right\}, \tag{63}$$

$$w_r^+ = \arg\max_{w\in\Delta_{k,s}} \left\{\langle \nabla Q_r(p_r),\, w - p_r\rangle - \tau_r D_{\mathrm{KL}}(w \,\|\, p_r)\right\} \tag{64}$$

have unique optimizers

$$w_{x,j}^+ \propto p_{x,j} \exp\left(\frac{(\nabla Q_x(p_x))_j}{\tau_x}\right), \tag{65}$$

$$w_{r,j}^+ \propto p_{r,j} \exp\left(\frac{(\nabla Q_r(p_r))_j}{\tau_r}\right). \tag{66}$$

Therefore, if the graph-induced priors satisfy $p_{x,j} \propto \xi_{x,j}^{\gamma_x/\tau_x}$ and $p_{r,j} \propto \xi_{r,j}^{\gamma_r/\tau_r}$, and if the local gradients are summarized by the branch utilities, then the proposed structure-aware adaptive weights coincide exactly with one entropic mirror-ascent / exponentiated-gradient step on those latent branch-quality functionals (Kivinen and Warmuth, 1997; Beck and Teboulle, 2003). This is a first-order relaxation statement, not a claim of a full global optimum for unconstrained LMB fusion.

A reduced-objective sensitivity view. Let

$$\Phi(w_x, w_r) := \min_{\{\beta_i\}} \mathcal{F}(w_x, w_r; \{\beta_i\}) \tag{67}$$

denote the aligned-LMB surrogate value function from Proposition C5. Under the same local differentiability assumptions used in standard envelope arguments (Milgrom and Segal, 2002), its branch sensitivities are

$$\frac{\partial \Phi}{\partial w_{x,j}} = \sum_i r_i^\star D_{\mathrm{KL}}(p_i^\star \,\|\, p_{i,j}), \tag{68}$$

$$\frac{\partial \Phi}{\partial w_{r,j}} = \sum_i d_{\mathrm{Ber}}(r_i^\star \,\|\, r_{i,j}). \tag{69}$$

Hence the exact reduced aligned-LMB objective penalizes spatial weights through existence-weighted spatial KL mismatch and penalizes existence weights through Bernoulli mismatch. This provides a concrete target for the retained utilities: covariance quality, realized link quality, and existence confidence should be read as stable surrogates for these exact reduced gradients, not as unrelated heuristic factors.

FID-FIA as a Bernoulli-branch separability surrogate. The reduced existence sensitivity above is a Bernoulli mismatch term. In labeled multi-target tracking, a large part of this mismatch is driven by whether the measurements provide enough information to distinguish target-generated returns from neighboring targets, clutter, or missed detections. The FID-FIA score measures exactly a local form of target-pair measurement separability through the Fisher metric induced by the sensor model. It is therefore a coherent surrogate for the reliability of the Bernoulli existence decision, even though it is not itself an exact Bernoulli KL gradient. This is the precise theoretical role used here: FID-FIA supplies an information-geometric separability utility for the existence branch, while covariance and link quality continue to supply the spatial and communication utilities.

C.5. Existence Confidence and Temporal Stabilization

The existence-confidence term uses

$$c(r) = |2r - 1|, \tag{70}$$

which is bounded in $[0, 1]$. This quantity is a bounded monotone surrogate of Bernoulli decisiveness because

$$\left|\log \frac{r}{1-r}\right| = \log \frac{1 + c(r)}{1 - c(r)} = 2\,\operatorname{artanh}(c(r)). \tag{71}$$

Therefore the existence-confidence score is a bounded monotone proxy for the magnitude of the Bernoulli natural parameter that appears in the exact existence branch (42).

The temporal stabilization step computes

$$w_k^{\mathrm{ema}} = \alpha w_{k-1} + (1-\alpha)\hat{w}_k. \tag{72}$$

Proposition C8. For any $\alpha \in [0,1]$, the EMA point w_k^{ema} is the unique minimizer of

$$\min_{w} \left\{ \alpha \|w - w_{k-1}\|_2^2 + (1-\alpha)\|w - \hat{w}_k\|_2^2 \right\}. \tag{73}$$

Thus the EMA step is an exact quadratic compromise between the previous weights and the instantaneous optimizer. The subsequent minimum-weight enforcement should still be interpreted as a pragmatic feasibility projection rather than as part of one single exact optimizer.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization. **Second Author**: Methodology, Supervision, Writing - review and editing, Project administration. **Third Author**: Supervision, Resources, Funding acquisition, Writing - review and editing.

Funding

This work was supported by [Funding Agency Name] under Grant No. [Grant Number].

Data availability

No real-world datasets were used in this study. The simulation data supporting the findings are synthetic and can be regenerated from the MATLAB/Octave scripts and deterministic seeds used in the experiments. The code and generated result logs will be made available by the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI Codex to assist with manuscript preparation and language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

Bandyopadhyay, S., Chung, S.J., 2018. Distributed bayesian filtering using logarithmic opinion pool for dynamic sensor networks. Automatica 97, 7–17. doi:10.1016/j.automatica.2018.07.013.

Bar-Shalom, Y., Li, X.R., Kirubarajan, T., 2001. Estimation with Applications to Tracking and Navigation. Wiley, New York.

Battistelli, G., Chisci, L., 2014. Kullback-leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability. Automatica 50, 707–718. doi:10.1016/j.automatica.2013.11.042.

Beck, A., Teboulle, M., 2003. Mirror descent and nonlinear projected subgradient methods for convex optimization. Operations Research Letters 31, 167–175. doi:10.1016/S0167-6377(02)00231-6.

Cao, C., Zhao, Y., 2026. Distributed heterogenous multi-sensor fusion based on labeled random finite set for multi-target state estimation using information geometry. Signal Processing 238, 110135. doi:10.1016/j.sigpro.2025.110135.

Chen, Z., Biggie, H., Ahmed, N., Julier, S., Heckman, C., 2023. Kalman filter auto-tuning through enforcing chi-squared normalized error distributions with bayesian optimization. doi:10.48550/arXiv.2306.07225, arXiv:2306.07225.

Gao, L., Battistelli, G., Chisci, L., 2020. Fusion of labeled RFS densities with minimum information loss. IEEE Transactions on Signal Processing 68, 5855–5868. doi:10.1109/TSP.2020.3028496.

Gao, L., Battistelli, G., Chisci, L., 2023. Resilient labeled multi-bernoulli fusion with peer-to-peer sensor network. Information Fusion 100, 101965. doi:10.1016/j.inffus.2023.101965.

Gostar, A.K., Rathnayake, T., Tennakoon, R.B., Bab-Hadiashar, A., Battistelli, G., Chisci, L., Hoseinnezhad, R., 2021. Centralized cooperative sensor fusion for dynamic sensor network with limited field-of-view via labeled multi-bernoulli filter. IEEE Transactions on Signal Processing 69, 878–891. doi:10.1109/TSP.2020.3048595.

Hlinka, O., Sluciak, O., Hlawatsch, F., Rupp, M., 2014. Distributed data fusion using iterative covariance intersection, in: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 1880–1884. doi:10.1109/ICASSP.2014.6853921.

Julier, S.J., Uhlmann, J.K., 1997. A non-divergent estimation algorithm in the presence of unknown correlations, in: Proceedings of the American Control Conference, pp. 2369–2373. doi:10.1109/ACC.1997.609105.

Kivinen, J., Warmuth, M.K., 1997. Exponentiated gradient versus gradient descent for linear predictors. Information and Computation 132, 1–63. doi:10.1006/inco.1996.2612.

Klupacs, J., Gostar, A.K., Bab-Hadiashar, A., Hoseinnezhad, R., 2023. An information-theoretic approach to complementary information fusion, in: Proceedings of the 12th International Conference on Control, Automation and Information Sciences, pp. 242–247. doi:10.1109/ICCAIS59597.2023.10382285.

Kuhn, H.W., 1955. The hungarian method for the assignment problem. Naval Research Logistics Quarterly 2, 83–97. doi:10.1002/nav.3800020109.

Li, S., Yi, W., Hoseinnezhad, R., Battistelli, G., Wang, B., Kong, L., 2018. Robust distributed fusion with labeled random finite sets. IEEE Transactions on Signal Processing 66, 278–293. doi:10.1109/TSP.2017.2760286.

Li, T., Corchado, J.M., Sun, S., 2017. On generalized covariance intersection for distributed PHD filtering and a simple but better alternative, in: Proceedings of the 20th International Conference on Information Fusion, pp. 1–8. doi:10.23919/ICIF.2017.8009732.

Li, T., Li, H., Li, G., Miao, H., 2026. Arithmetic average density fusion–part V: Fisher information weighted fusion of multi-rate filters. TechRxiv preprint. doi:10.36227/techrxiv.176782063.38593428/v1. posted January 7, 2026.

Li, T., Yan, R., Da, K., Fan, H., 2024. Arithmetic average density fusion–part III: Heterogeneous unlabeled and labeled RFS filter fusion. IEEE Transactions on Aerospace and Electronic Systems 60, 1023–1034. doi:10.1109/TAES.2023.3334223.

Mahler, R.P.S., 2000. Optimal/robust distributed data fusion: A unified approach, in: Signal Processing, Sensor Fusion, and Target Recognition IX, pp. 128–138. doi:10.1117/12.395064.

Mahler, R.P.S., 2003. Multitarget bayes filtering via first-order multitarget moments. IEEE Transactions on Aerospace and Electronic Systems 39, 1152–1178. doi:10.1109/TAES.2003.1261119.

Mahler, R.P.S., 2007. Statistical Multisource-Multitarget Information Fusion. Artech House, Boston, MA.

Mahler, R.P.S., 2019. Exact closed-form multitarget bayes filters. Sensors 19, 2818. doi:10.3390/s19122818.

Milgrom, P., Segal, I., 2002. Envelope theorems for arbitrary choice sets. Econometrica 70, 583–601. doi:10.1111/1468-0262.00296.

Noack, B., Sijs, J., Reinhardt, M., Hanebeck, U.D., 2017. Decentralized data fusion with inverse covariance intersection. Automatica 79, 35–41. doi:10.1016/j.automatica.2017.01.019.

Rahmathullah, A.S., Garcia-Fernandez, A.F., Svensson, L., 2017. Generalized optimal sub-pattern assignment metric, in: Proceedings of the 20th International Conference on Information Fusion, pp. 1–8. doi:10.23919/ICIF.2017.8009645.

Reuter, S., Vo, B.T., Vo, B.N., Dietmayer, K., 2014. The labeled multi-bernoulli filter. IEEE Transactions on Signal Processing 62, 3246–3260. doi:10.1109/TSP.2014.2323064.

Ristic, B., Vo, B.N., Clark, D., Vo, B.T., 2011. A metric for performance evaluation of multi-target tracking algorithms. IEEE Transactions on Signal Processing 59, 3452–3457. doi:10.1109/TSP.2011.2140111.

Schuhmacher, D., Vo, B.T., Vo, B.N., 2008. A consistent metric for performance evaluation of multi-object filters. IEEE Transactions on Signal Processing 56, 3447–3457. doi:10.1109/TSP.2008.920469.

Shen, K., Dong, P., Jing, Z., Leung, H., 2022. Consensus-based labeled multi-bernoulli filter for multitarget tracking in distributed sensor network. IEEE Transactions on Cybernetics 52, 12722–12733. doi:10.1109/TCYB.2021.3087521.

Uhlmann, J.K., 2003. Covariance consistency methods for fault-tolerant distributed data fusion. Information Fusion 4, 201–215. doi:10.1016/S1566-2535(03)00036-8.

Üney, M., Clark, D.E., Julier, S.J., 2013. Distributed fusion of PHD filters via exponential mixture densities. IEEE Journal of Selected Topics in Signal Processing 7, 521–531. doi:10.1109/JSTSP.2013.2257162.

Üney, M., Houssineau, J., Delande, E., Julier, S.J., Clark, D.E., 2019. Fusion of finite-set distributions: Pointwise consistency and global cardinality. IEEE Transactions on Aerospace and Electronic Systems 55, 2759–2773. doi:10.1109/TAES.2019.2893083.

Vo, B.N., Vo, B.T., Beard, M., 2019. Multi-sensor multi-object tracking with the generalized labeled multi-bernoulli filter. IEEE Transactions on Signal Processing 67, 5952–5967. doi:10.1109/TSP.2019.2946023.

Vo, B.N., Vo, B.T., Nguyen, T.T.D., Shim, C., 2024. An overview of multi-object estimation via labeled random finite set. IEEE Transactions on Signal Processing 72, 4888–4917. doi:10.1109/TSP.2024.3472068.

Vo, B.N., Vo, B.T., Phung, D., 2014. Labeled random finite sets and the Bayes multi-target tracking filter. IEEE Transactions on Signal Processing 62, 6554–6567. doi:10.1109/TSP.2014.2364014.

Vo, B.T., Ma, W.K., 2006. The gaussian mixture probability hypothesis density filter. IEEE Transactions on Signal Processing 54, 4091–4104. doi:10.1109/TSP.2006.881190.

Wang, B., Yi, W., Hoseinnezhad, R., Li, S., Kong, L., Yang, X., 2017. Distributed fusion with multi-bernoulli filter based on generalized covariance intersection. IEEE Transactions on Signal Processing 65, 242–255. doi:10.1109/TSP.2016.2617825.

Wang, X., Gostar, A.K., Rathnayake, T., Xu, B., Bab-Hadiashar, A., Hoseinnezhad, R., 2018. Centralized multiple-view sensor fusion using labeled multi-bernoulli filters. Signal Processing 150, 75–84. doi:10.1016/j.sigpro.2018.04.010.

Williams, J.L., Lau, R.A., 2014. Approximate evaluation of marginal association probabilities with belief propagation. IEEE Transactions on Aerospace and Electronic Systems 50, 2942–2959. doi:10.1109/TAES.2014.120568.

Xiao, L., Boyd, S., Lall, S., 2005. A scheme for robust distributed sensor fusion based on average consensus, in: Proceedings of the 4th International Symposium on Information Processing in Sensor Networks, pp. 63–70. doi:10.1109/IPSN.2005.1440896.

Zhang, W.A., Chen, B., Song, H., Yu, L., 2016. Distributed Fusion Estimation for Sensor Networks with Communication Constraints. Springer, Singapore. doi:10.1007/978-981-10-0795-8.

Zhang, X.H., 2020. An optimal data fusion algorithm in the presence of unknown cross covariances. IEEE Transactions on Automatic Control 65, 1226–1233. doi:10.1109/TAC.2019.2925500.

Zheng, L., Shi, L., Yang, F., Cai, Y., 2026. Distributed weighted fusion of RFS densities for peer-to-peer sensor networks. Information Fusion 129, 104060. doi:10.1016/j.inffus.2025.104060.